Institut National de Statistique
et d'Économie Appliquée

From Statistics to Machine Learning

Class notes

FSML

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Prologue
Data, models, and learning

This prologue sketches the arc of the course: from a table of numbers to a model of the process that produced it, and from models of processes to machines that predict. Each idea returns later, in this course or in its follow-up on time series and uncertainty.

1 Data is multivariate

A shop tracks its loyalty-card customers. Its database holds one row per customer:

	visits / month	spend / month (Dh)	days since last visit
customer 1	11	480	3
customer 2	7	310	8
customer 3	16	690	1
customer 4	12	520	4
customer 5	5	260	12

Each row collects several quantities measured simultaneously on the same customer: an *observation* is not a number but a vector,

$$\mathbf{x}_i = (x_{i1}, \ldots, x_{id})^\top \in \mathbb{R}^d, \qquad i = 1, \ldots, n, \tag{1}$$

and the table stacks the n observations as the rows of a matrix in $\mathbb{R}^{n \times d}$ (here $d = 3$). The notation in (1) fixes a convention used throughout: bold letters ($\mathbf{x}_i$) denote vectors, plain italic letters denote one-dimensional quantities, such as the components x_{ij}. Frequent visitors spend more: the columns move together, and the way they move together is often the information sought. Segmenting the customers into profiles, or spotting the odd ones, is done on combinations of columns, not on any column alone.

Tables of this kind are everywhere. A bank records card transactions (amount, hour, merchant category, minutes since the previous transaction), and fraud lives in the combination: a large electronics purchase at 3 a.m., seconds after the previous transaction, where each feature taken alone is unremarkable. A sales team keeps one row per day (price, promotion, units sold); an insurer one row per claim (amount, delay, category); a biology lab one row per patient, with thousands of gene-expression levels that co-vary in pathways. In every case the natural unit is the row $\mathbf{x}_i \in \mathbb{R}^d$, with d ranging from a handful to many thousands.

2 From columns to a joint model

First-year statistics equips us to model a single column: draw its histogram, fit a distribution, compute quantiles. Nothing prevents applying that recipe to the shop's table one column at a time, and figure 1 shows the result for its 400 customers: three histograms, one per column, each perfectly legible on its own. Visits center near 12 per month, spend near 530 Dh, and the days since the last visit pile up early (84% of customers came within the last week), with a long right tail of absentees. For a question that concerns one column alone (what fraction of customers visits more than 15 times a month?), these plots are all that is needed.

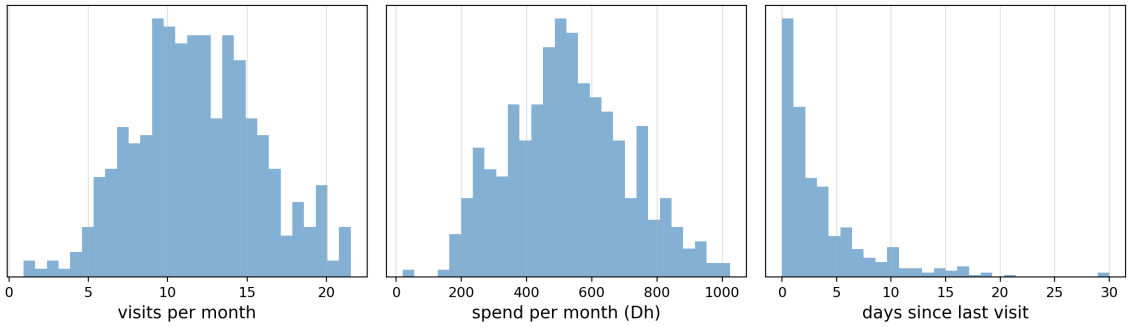


Figure 1: The shop's table, one column at a time: a histogram per column for the 400 customers. Each marginal distribution is legible on its own; nothing in the three plots reveals how the columns move together.

The shop's questions, however, concern the columns *jointly*, and the column-by-column view is blind to precisely that. Figure 2 plots the same customers by the first two columns together, visits per month against monthly spend. The two columns are strongly dependent, since customers who come often spend more in total. The customer marked in red makes 17 visits (more than 90% of customers, but nowhere near a record) and spends 300 Dh (more than 12% of customers, again well inside the observed range). Examined column by column, this customer triggers nothing: both red lines fall well inside their histograms, the very histograms of figure 1. Examined as a *pair*, the same customer sits far outside the cloud: someone who visits every other day yet buys almost nothing, a combination the process essentially never produces. Whether this signals fraud, a malfunctioning loyalty card, or a very patient window-shopper is a business question; that it must be *flagged* at all is visible only jointly.

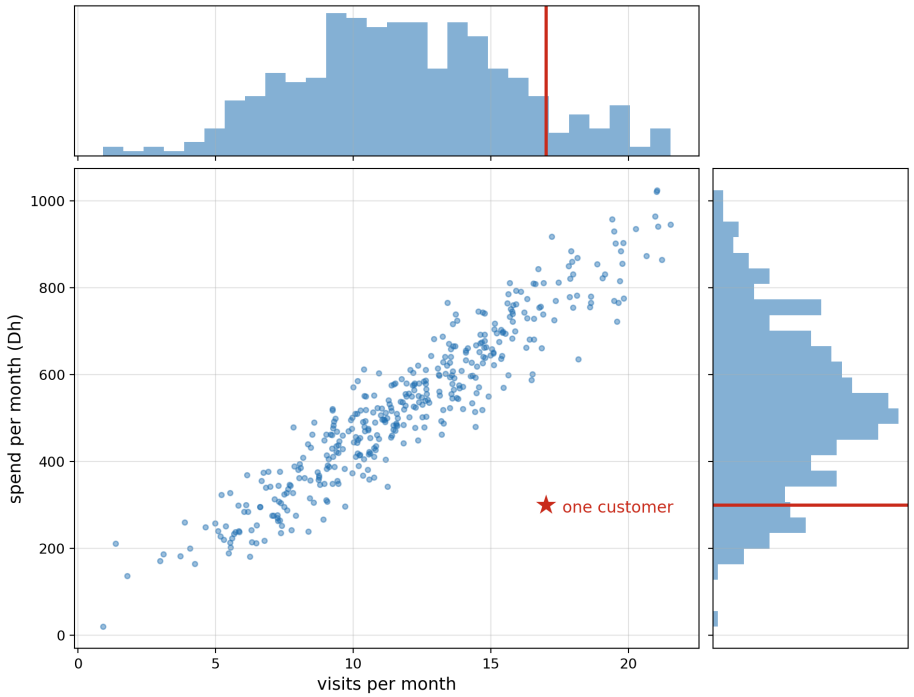


Figure 2: Four hundred customers described by two columns. The customer marked in red is unremarkable in each marginal histogram (red lines) yet far from the cloud as a pair: univariate screening, column by column, cannot flag what the joint view makes obvious.

In general, the marginal distributions of the columns do not determine the distribution of the

row: infinitely many joint distributions share the same d marginals and differ only in how the columns are tied together. The row must therefore be treated as a single object (the vector $\mathbf{x}_i$ of (1)), and the object to model is its *joint* distribution.

A table of n rows is, by itself, only a record of the past. To *model* it is to make a decision: to treat the rows as the output of a data-generating process, in the simplest and most common form,

$$\mathbf{x}_1, \dots, \mathbf{x}_n \overset{\text{i.i.d.}}{\sim} P, \tag{2}$$

independent draws of a *random vector* $\mathbf{X}$ with a single unknown probability distribution P on $\mathbb{R}^d$, identically distributed because the process is assumed stable across rows. The abbreviation i.i.d. (*independent and identically distributed*) will be used constantly in this course; when P admits a density on $\mathbb{R}^d$, we write it f.

The decision is consequential because a distribution can be interrogated in ways a table cannot. If P were known, we could compute the probability $\mathbb{P}(\mathbf{X} \in A)$ of any future event A (a transaction above 5 000 Dh tonight), and summarize $n \times d$ recorded numbers by the few parameters that pin down P. Five uses recur throughout the course: describe the *typical* rows; read the *dependence* between columns; flag any new row as *anomalous* under the process (the red customer of figure 2 would receive a near-zero density); fill in a *missing* entry from the observed ones, using that dependence; and *generate* synthetic rows indistinguishable in law from the real ones. A single distribution serves all five purposes.

The first part of this course follows this program: choose a family of candidate distributions on $\mathbb{R}^d$, use the table to determine which member of the family the process resembles, and then interrogate it. Whether the assumptions behind the program (rows independent, process stable, family adequate) hold is examined from chapter 1 onward.

3 Predicting one column from the others

Modeling the full joint distribution is the most complete ambition, and the most expensive one. Often the question at hand is narrower. One column has a special status (the quantity to be predicted) and the others are context: units sold (Y) given price and promotion ($\mathbf{X}$); fraudulent or not (Y) given the transaction's features ($\mathbf{X}$); a patient's outcome (Y) given the covariates ($\mathbf{X}$). The question is not "how is the table distributed?" but "what will Y be when $\mathbf{X} = \mathbf{x}$?"

The joint distribution of the pair splits into two factors: the distribution of $\mathbf{X}$ itself, and the conditional distribution of Y given $\mathbf{X}$,

$$\underbrace{f(\mathbf{x}, y)}_{\text{joint}} = \underbrace{f_{\mathbf{X}}(\mathbf{x})}_{\text{context}} \times \underbrace{f_{Y|\mathbf{X}}(y \mid \mathbf{x})}_{\text{what we predict}} . \tag{3}$$

Prediction needs only the second factor. How prices came to be set is irrelevant to predicting demand on a day whose price is observed; the first factor can be left unmodeled entirely. (What demand would be if we *set* the price ourselves is a different question, a causal one, which this factorization does not answer.) This is a substantial economy: instead of a distribution on $\mathbb{R}^d$, one conditional distribution on $\mathbb{R}$, indexed by the context.

Modeling $Y \mid \mathbf{X}$ under explicit assumptions is *regression*, and its simplest instance is already familiar from first year: the linear model, where Y given $\mathbf{X} = \mathbf{x}$ is Gaussian with mean $\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}$ and constant variance σ^2. Chapter 4 re-examines that familiar object as a statistical model, with likelihoods, tests, and honest uncertainty; the second part of the course then deliberately loosens its assumptions.

4 From distributions to functions

Sometimes even the conditional distribution resists credible specification. The dependence of Y on twenty interacting covariates is nonlinear in ways no one can write down in advance; the Gaussian

noise assumption is questionable; every added assumption is one more thing that can be false. The stance of machine learning is to retreat further: stop specifying distributions altogether, and learn a bare function $f : \mathbf{x} \mapsto \hat{y}$ directly from the table, judged by a single criterion (how well it predicts rows it has never seen). Where the statistician asks "is the model right?", the learner asks only "does it generalize?".

The gain is flexibility: freed from distributional form, the function f can be a decision tree, an ensemble of hundreds of trees, a boosted committee (the machinery of the last chapter of this course), or a deep network, beyond it.

5 The price: uncertainty lost, and bought back

A conditional distribution answers "what will Y be?" with a full range of possibilities and their probabilities. A function answers with one number. Figure 3 shows the retreat on a single dataset: the monthly spend of a store's customers against their visit frequency, one point per customer. Panel (a) models the pair completely; panel (b) keeps only the conditional distributions of spend given visits, whose spread visibly grows with the visit frequency; panel (c) is the pure learner's output, a curve: accurate on average, and silent about everything else. This matters whenever the decision depends on more than the average: what spend a new ten-visit customer will bring nine times out of ten, whether next month's figure could plausibly be double the prediction, how much trust a single number deserves.

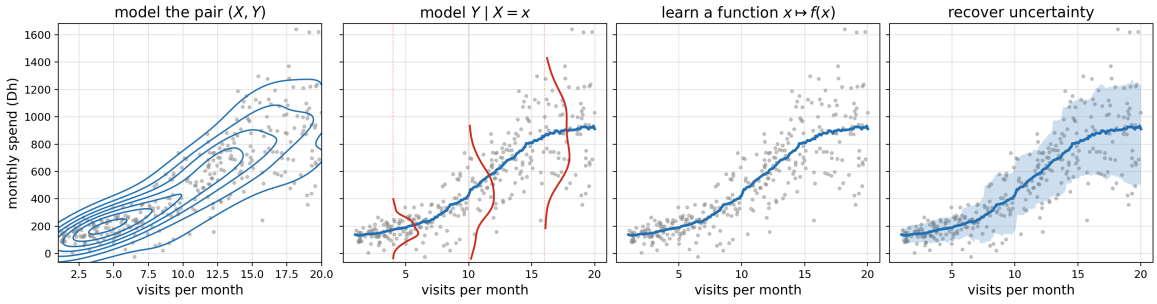


Figure 3: One dataset, four stances. (a) Model the joint distribution of the pair. (b) Model only the conditional distribution of spend given visits (red: three of its slices; note the growing spread). (c) Learn a bare function: one number per visit frequency. (d) Recover uncertainty around the function: a band containing 80% of the outcomes at each visit frequency.

Panel (d) shows what modern practice adds back: a band around the prediction calibrated so that, at each visit frequency, it contains a stated fraction of the outcomes. Two routes lead toward such bands without reinstating a full probabilistic model. *Quantile regression* learns, instead of the average, the conditional quantiles of Y directly: the edges of the band as functions of $\mathbf{x}$. *Conformal prediction* wraps any learned predictor, however opaque, in intervals whose coverage *on average over new rows* is guaranteed by the i.i.d. assumption alone; coverage at each value of $\mathbf{x}$ separately is not guaranteed. *Bayesian modeling* goes the other way: it keeps distributions everywhere, placing them on the parameters themselves so that predictions inherit a spread. Prediction uncertainty is taken up in earnest in the follow-up course, *Advanced Time Series* (ATS), through quantile regression and conformal prediction.

6 When the rows have an order: beyond i.i.d.

Everything so far treats the rows as exchangeable: shuffle the table and nothing of interest changes. For many tables this is right: customers in a database have no natural order. But rows are often *dated*, and then the order may carry the very structure we care about. Figure 4 shows

a year of the store's daily sales twice: in calendar order, and shuffled. An i.i.d. model sees the two panels identically (it retains the histogram of values and nothing else), yet the panels differ in the way that matters most: on the left, tomorrow resembles today, the high season swells and recedes, and forecasting is possible; on the right, knowing today says nothing about tomorrow.

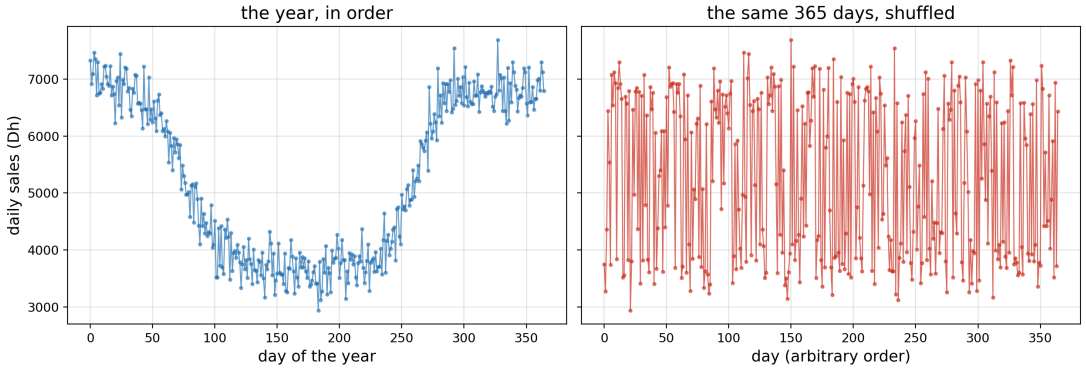


Figure 4: A year of the store's daily sales, in calendar order (left) and shuffled (right). Both tables have exactly the same 365 values, so an i.i.d. model cannot distinguish them; the forecastable structure lives entirely in the order.

Modeling dependence *across* rows (trends, seasonality, autocorrelation) is time-series modeling, and first year devoted an entire course to it: SARIMA models, built and read in the econometrician's spirit, where the order of the rows is the very subject of the model. This course does not rerun that material: ordered rows are the subject of the follow-up course, *Advanced Time Series* (ATS), which builds on both stances of this prologue. It recaps the classical models (exponential smoothing, SARIMA with exogenous covariates, where the forecast of a series is helped by other columns measured alongside it), then takes the learner's side: an autoregression rewritten as a supervised problem, so that the machine-learning models of this course (regularized regression, trees, gradient boosting) become forecasters fed with the order as context (yesterday's value, last week's, the calendar). Evaluation and pipelines are rebuilt for dated data: baselines to beat, rolling-origin backtesting instead of random splits, and features built only from the information available when the forecast is made.

This course, then, works under the i.i.d. assumption, and for each dataset met along the way, whether exchangeability is a reasonable approximation is part of the modeling judgment, not a formality.

7 A map of the course

The course has two parts. **Part 1** carries out the programs of sections 2 and 3: chapter 1 builds the workhorse family of distributions on $\mathbb{R}^d$, the multivariate Gaussian, and confronts the cost of estimating it; chapter 2 stacks several Gaussians into mixtures to describe populations with hidden subgroups, and faces the question every enlarged family forces: how to choose among models of different complexity; chapter 3 rebuilds hypothesis testing, the art of turning probabilities computed under a model into evidence about a claim; and chapter 4, following section 3, re-reads regression as a statistical model, where assumptions yield tests and confidence statements.

Part 2 follows section 4: from models to predictors (chapter 5), where generalization replaces inference and we measure what is lost in doing so; the machine-learning pipeline (chapter 6), where predictors meet raw tables and honest evaluation; and trees and their ensembles (chapter 7), where flexibility peaks. Recovering uncertainty (section 5) and ordered rows (section 6) are left to the ATS course.

The course aims to give the judgment needed to choose, in front of a real table, between the two approaches: assumptions chosen with care, which turn scarce data into strong statements,

or abundant data, which replaces assumptions.

Part I

Multivariate statistical modeling

Chapter 1
Statistical models and the multivariate Gaussian

Most datasets record several quantities at once: the mathematics and physics scores of each candidate in an entrance exam, blood pressure *and* cholesterol for each patient, the amount, hour and merchant category of each card transaction, the expression of thousands of genes in one cell. The prologue turned this observation into a program: treat the rows of such a table as independent draws of a random vector, fix a family of candidate distributions, let the data select a member, then interrogate it. This chapter carries out the program once, in full, with the central family for multivariate data: the multivariate normal, or Gaussian, distribution. By the end of the chapter we will know what this model can do (describe dependence, calibrate anomalies, impute missing values, quantify its own uncertainty), what it costs (a covariance matrix whose size grows quadratically with the dimension), and where it breaks (data with groups). That failure motivates the next chapter.

Throughout, the prologue's notational convention holds: bold letters denote vectors, plain italic letters denote one-dimensional quantities. $\mathbf{X}$ is a random vector in $\mathbb{R}^d$ with components $X_1, \dots, X_d$, and Σ its covariance matrix. Two toolboxes are used constantly and collected in the appendices: the spectral decomposition of symmetric matrices (appendix A) and the basic calculus of random vectors (expectations, covariance matrices, characteristic functions; appendix B). If you have not manipulated these objects recently, read the appendices first: every argument in this chapter uses them.

1.1 Gaussian vectors: definition, density, geometry

The chapter works with the kind of table the course opened with: one row per observation, one column per measured quantity; in generic notation, d variables $X_1, \dots, X_d$ measured on each of n observation units,

	X_1	X_2	$\cdots$	X_d
observation 1	x_{11}	x_{12}	$\cdots$	x_{1d}
observation 2	x_{21}	x_{22}	$\cdots$	x_{2d}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
observation n	x_{n1}	x_{n2}	$\cdots$	x_{nd}

so each row is a point $\mathbf{x}_i \in \mathbb{R}^d$: a customer, a patient, a transaction, an exam candidate, whatever the unit of observation is. The prologue laid out what a distribution P fitted to such rows buys (typical rows, dependence between columns, calibrated anomalies, missing entries filled in, synthetic data) and how the impossible task of guessing a whole distribution is made tractable: by fixing a family of candidate distributions in advance. One definition, met in your first-year courses in its univariate form, makes this precise.

Reminder

(*Parametric statistical model.*) A *statistical model* for observations $\mathbf{x}_1, \dots, \mathbf{x}_n$ is a family of probability distributions

$$\mathcal{P} = \{P_\theta,\ \theta \in \Theta\},$$

together with the assumption that the observations are independent draws from P_{θ^*} for some unknown $\theta^* \in \Theta$. The model is *parametric* when $\Theta \subset \mathbb{R}^k$ for a finite k.

A parametric model is a drastic compression: instead of an unknown function (a full distribution on $\mathbb{R}^d$), the unknown is a vector of k numbers. Every request made of the table becomes a question about θ^*, and "learning the distribution" becomes "estimating k parameters from n rows", a competition between k and n that will be a recurring theme of this course, starting in section 1.6.2.

It remains to choose the family, and this chapter develops the simplest serious candidate for multivariate data: the multivariate analogue of the distribution every first-year course reaches for first.

From one dimension to d. On the real line, the Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$, the workhorse of your first-year statistics courses, has density $f(x) = (2\pi\sigma^2)^{-1/2}\, e^{-(x-\mu)^2/(2\sigma^2)}$: one center μ, one spread σ. How should one define its analogue for a random vector $\mathbf{X} = (X_1, \dots, X_d)^\top$ (a distribution for whole rows of the table rather than single numbers)? A first guess is to require each component X_i to be Gaussian. This guess is *wrong*, and seeing why it fails leads to the correct definition.

Take $X \sim \mathcal{N}(0,1)$, flip an independent fair coin $\varepsilon \in \{-1, +1\}$, and set $Y = \varepsilon X$. Flipping the sign of a symmetric random variable does not change its distribution, so Y is also a standard Gaussian. But now consider the pair $\mathbf{Z} = (X, Y)$ and look at the sum of its components:

$$X + Y = (1+\varepsilon)X = \begin{cases} 2X & \text{with probability } 1/2, \\ 0 & \text{with probability } 1/2. \end{cases}$$

Half of the mass of $X+Y$ sits exactly at 0: this distribution has an atom, so it cannot be Gaussian. The pair $\mathbf{Z}$ has two perfectly Gaussian coordinates, yet something un-Gaussian happens as soon as we combine them. Componentwise Gaussianity does not constrain the joint behavior, which is the object of this chapter. The correct definition demands Gaussianity of every one-dimensional projection of the vector:

Definition 1.1.1

A random vector $\mathbf{X}$ with mean $\boldsymbol{\mu}$ and covariance Σ is a *Gaussian vector*, written $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if every linear combination of its components is univariate Gaussian:

$$\forall \boldsymbol{\alpha} \in \mathbb{R}^d, \qquad \boldsymbol{\alpha}^\top \mathbf{X} = \sum_{i=1}^{d} \alpha_i X_i \;\sim\; \mathcal{N}\Big(\boldsymbol{\alpha}^\top \boldsymbol{\mu},\; \boldsymbol{\alpha}^\top \Sigma\, \boldsymbol{\alpha}\Big). \tag{4}$$

The parameters of $\boldsymbol{\alpha}^\top \mathbf{X}$ are forced: its mean is $\boldsymbol{\alpha}^\top \boldsymbol{\mu}$ by linearity of expectation, and its variance is $\boldsymbol{\alpha}^\top \Sigma\, \boldsymbol{\alpha}$ by the identity $\mathbb{V}(\boldsymbol{\alpha}^\top \mathbf{X}) = \boldsymbol{\alpha}^\top \mathbb{V}(\mathbf{X})\, \boldsymbol{\alpha}$ (appendix B). The definition is legitimate (it pins down one distribution) thanks to the Cramér–Wold theorem (appendix B): the distribution of a random vector is entirely determined by the distributions of its one-dimensional projections $\mathbf{t}^\top \mathbf{X}$. In our counter-example, $\mathbf{Z} = (X, \varepsilon X)$ fails the test at $\boldsymbol{\alpha} = (1,1)^\top$: it is *not* a Gaussian vector, although both its marginals are.

The density. When Σ is invertible, the Gaussian vector admits a density, which is the direct generalization of the 1D formula: the squared standardized deviation $(x-\mu)^2/\sigma^2$ becomes a quadratic form in the inverse covariance.

Property 1.1.1

If $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with Σ invertible, then $\mathbf{X}$ has density

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}\, |\Sigma|^{1/2}} \exp\Big(-\tfrac{1}{2}\,(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})\Big), \qquad \mathbf{x} \in \mathbb{R}^d. \tag{5}$$

Two sanity checks help make this formula familiar. For $d = 1$, $\Sigma = \sigma^2$ and we recover the 1D density. For $\Sigma = \mathrm{I}_d$ and $\boldsymbol{\mu} = 0$, the density factorizes as $\prod_i (2\pi)^{-1/2} e^{-x_i^2/2}$: independent standard

Gaussian coordinates. The general case is, as we will see in section 1.3, nothing but a linear deformation of this product case.

Geometry: where does the mass sit? The density is constant on the level sets

$$\mathcal{E}_r = \left\{ \mathbf{x} \in \mathbb{R}^d : \ (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = r^2 \right\}.$$

To read this equation, diagonalize the covariance: $\Sigma = P \Lambda P^\top$ with P orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$ (appendix A). In the rotated, centered coordinates $\mathbf{y} = P^\top (\mathbf{x} - \boldsymbol{\mu})$, the equation becomes

$$\sum_{i=1}^{d} \frac{y_i^2}{\lambda_i} = r^2,$$

the equation of an ellipsoid with axes aligned with the coordinate axes and semi-axis lengths $r\sqrt{\lambda_i}$. Undoing the rotation, $\mathcal{E}_r$ is an ellipsoid centered at $\boldsymbol{\mu}$ whose principal axes are the eigenvectors $\mathbf{e}_i$ of Σ and whose semi-axes have lengths $r\sqrt{\lambda_i}$ (figure 1.1). A common trap: since Σ^{-1} appears in the density, one is tempted to read the geometry off Σ^{-1}. But inverting a symmetric matrix keeps the eigenvectors and inverts the eigenvalues, so the *largest* eigenvalue of Σ (the direction of largest variance) gives the *longest* axis of the ellipsoid, as it should.

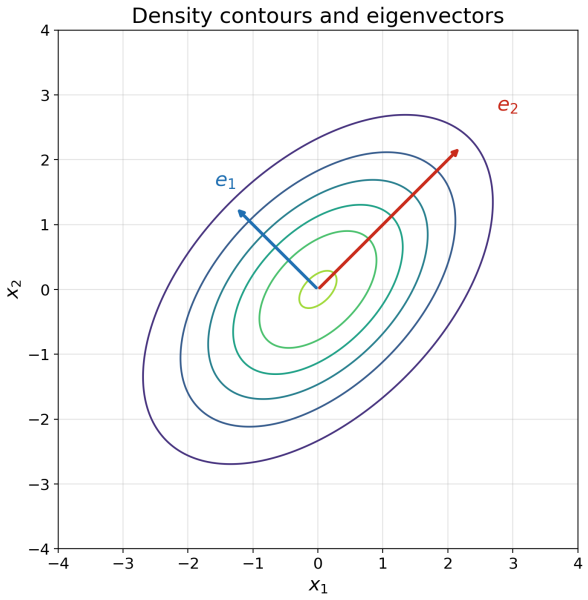


Figure 1.1: Density contours of $\mathcal{N}(0, \Sigma)$, with the eigenvectors of Σ (here $\Sigma_{11} = \Sigma_{22} = 2$, $\Sigma_{12} = 1$). The long axis follows the top eigenvector of Σ.

Degenerate Gaussians. Definition 1.1.1 does not require Σ to be invertible; only the density formula does. What happens when Σ is singular? Some direction $\boldsymbol{\alpha} \neq 0$ satisfies $\mathbb{V}(\boldsymbol{\alpha}^\top \mathbf{X}) = \boldsymbol{\alpha}^\top \Sigma \, \boldsymbol{\alpha} = 0$: the vector lies, with probability one, in the affine subspace $\{\mathbf{x} : \boldsymbol{\alpha}^\top (\mathbf{x} - \boldsymbol{\mu}) = 0\}$. The distribution then concentrates on $\boldsymbol{\mu} + \mathrm{Im}(\Sigma)$, a subspace of dimension $\mathrm{rank}(\Sigma) < d$, and no density on $\mathbb{R}^d$ exists. This is not a pathology reserved to textbooks: figure 1.2 shows samples from a bivariate Gaussian with unit variances and correlation ρ. As $\rho \to 1$ the ellipse flattens, and at $\rho = 1$ the cloud collapses onto the line $x_2 = x_1$: a legitimate Gaussian vector living on a one-dimensional subspace. Singular covariance matrices return in section 1.6.2, produced there by estimation itself when the dimension approaches the sample size.

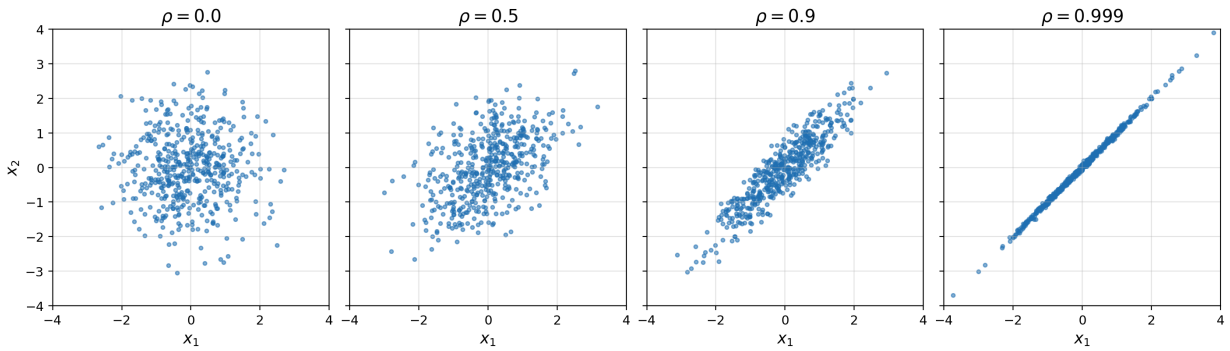


Figure 1.2: Samples of $\mathcal{N}(0, \Sigma)$ with unit variances and correlation ρ: the cloud flattens onto a line as $\rho \to 1$.

1.2 Marginals and independence

Partition the vector into two blocks: pick $1 \le r < d$ and write

$$\mathbf{X} = (\underbrace{X_1, \dots, X_r}_{\mathbf{X}_a \in \mathbb{R}^r}, \underbrace{X_{r+1}, \dots, X_d}_{\mathbf{X}_b \in \mathbb{R}^{d-r}})^\top.$$

Each block is itself a random *vector*: $\mathbf{X}_a$ of dimension r, $\mathbf{X}_b$ of dimension $d-r$, the two dimensions summing to d; blocks of size one are just the special case $r = 1$ or $r = d-1$. The mean and covariance split accordingly:

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix},$$

where $\boldsymbol{\mu}_a \in \mathbb{R}^r$, $\Sigma_{aa} \in \mathbb{R}^{r \times r}$ and $\Sigma_{bb} \in \mathbb{R}^{(d-r)\times(d-r)}$ are the within-block means and covariances, while the rectangular block $\Sigma_{ab} = \Sigma_{ba}^\top \in \mathbb{R}^{r \times (d-r)}$ collects all the covariances *between* a coordinate of $\mathbf{X}_a$ and a coordinate of $\mathbf{X}_b$. Two natural questions arise immediately. What is the distribution of $\mathbf{X}_a$ alone (the marginal)? And when are $\mathbf{X}_a$ and $\mathbf{X}_b$ independent? For Gaussian vectors both have closed-form answers, read off directly from the blocks. A third question, what *observing* $\mathbf{X}_b$ teaches us about $\mathbf{X}_a$, has an equally explicit answer, and it is the key to filling in missing entries.

Marginals. A linear combination of the components of $\mathbf{X}_a$ is in particular a linear combination of the components of $\mathbf{X}$, so it is univariate Gaussian: by definition 1.1.1, $\mathbf{X}_a$ is itself a Gaussian vector, and reading off its mean and covariance gives $\mathbf{X}_a \sim \mathcal{N}(\boldsymbol{\mu}_a, \Sigma_{aa})$. Marginalizing a Gaussian is as simple as extracting sub-blocks. Figure 1.3 shows the phenomenon on simulated data: however correlated the cloud, its shadow on each axis is a Gaussian histogram.

Independence. The most efficient tool here is the characteristic function, which for Gaussian vectors has a remarkably simple form.

Property 1.2.1

For $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$,

$$\phi_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}\left[e^{i\mathbf{t}^\top \mathbf{X}}\right] = \exp\left(i\,\mathbf{t}^\top \boldsymbol{\mu} - \tfrac{1}{2}\,\mathbf{t}^\top \Sigma\,\mathbf{t}\right), \qquad \mathbf{t} \in \mathbb{R}^d. \tag{6}$$

Write $\mathbf{t} = (\mathbf{t}_a, \mathbf{t}_b)$ and expand the quadratic form by blocks:

$$\mathbf{t}^\top \Sigma\,\mathbf{t} = \mathbf{t}_a^\top \Sigma_{aa} \mathbf{t}_a + \mathbf{t}_b^\top \Sigma_{bb} \mathbf{t}_b + 2\,\mathbf{t}_a^\top \Sigma_{ab}\,\mathbf{t}_b.$$

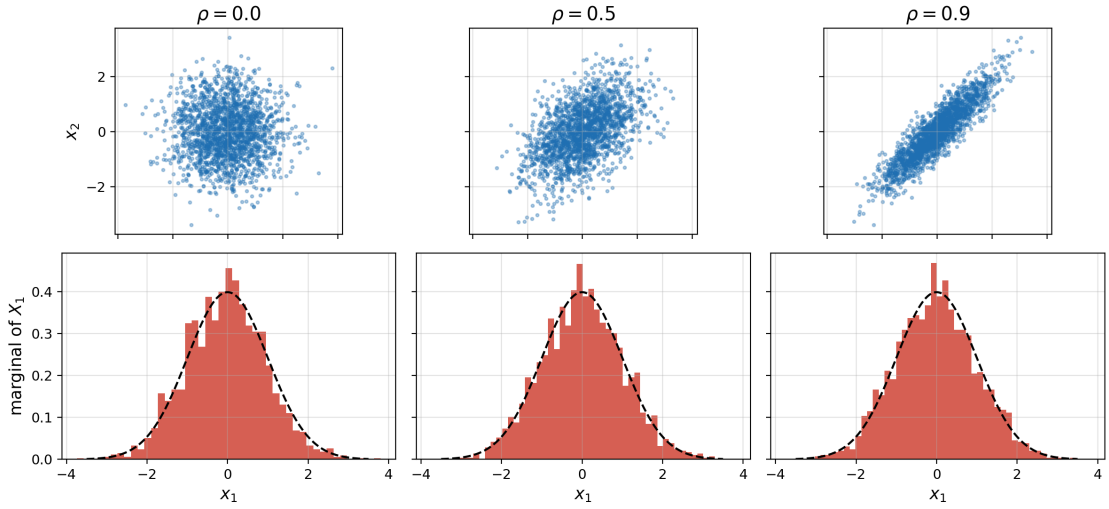


Figure 1.3: Correlated Gaussian samples (top row) and the marginal distribution of X_1 (bottom row): Gaussian at every correlation level.

If $\Sigma_{ab} = 0$, the cross term vanishes and $\phi_{\mathbf{X}}(\mathbf{t})$ factorizes as $\phi_{\mathbf{X}_a}(\mathbf{t}_a)\,\phi_{\mathbf{X}_b}(\mathbf{t}_b)$, which characterizes independence (appendix B); the converse is immediate since independence forces zero covariance. We have proved:

Theorem 1.2.1

For a Gaussian vector, $\mathbf{X}_a \perp\!\!\!\perp \mathbf{X}_b \iff \Sigma_{ab} = 0$: independence is equivalent to zero correlation.

This equivalence is a property of jointly Gaussian vectors, not of Gaussian marginals (the pair $(X, \varepsilon X)$ of section 1.1 has uncorrelated standard Gaussian coordinates, yet $|\varepsilon X| = |X|$), and a common source of mistakes outside them. In general, zero correlation is much weaker than independence: take $X_1 \sim \mathcal{N}(0,1)$ and $X_2 = X_1^2$. Then $\mathrm{Cov}(X_1, X_2) = \mathbb{E}(X_1^3) = 0$ by symmetry, yet X_2 is a deterministic function of X_1. Correlation measures only *linear* association.

Conditionals. The third question, what observing $\mathbf{X}_b$ tells us about $\mathbf{X}_a$, is answered by the following theorem.

Theorem 1.2.2 (Conditional distribution)

If Σ_{bb} is invertible, then

$$\mathbf{X}_a \mid \mathbf{X}_b = \mathbf{x}_b \;\sim\; \mathcal{N}\Big(\boldsymbol{\mu}_a + \Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{x}_b - \boldsymbol{\mu}_b),\; \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}\Big)\,. \tag{7}$$

Proof. The trick is to remove from $\mathbf{X}_a$ the part of it that $\mathbf{X}_b$ "explains". Set $\mathbf{W} = \mathbf{X}_a - \Sigma_{ab}\Sigma_{bb}^{-1}\mathbf{X}_b$. The pair $(\mathbf{W}, \mathbf{X}_b)$ is an affine transformation of $\mathbf{X}$, hence Gaussian (property 1.3.1 of section 1.3), and

$$\mathrm{Cov}(\mathbf{W}, \mathbf{X}_b) = \Sigma_{ab} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{bb} = 0,$$

so by theorem 1.2.1, $\mathbf{W}$ is independent of $\mathbf{X}_b$. Now write $\mathbf{X}_a = \mathbf{W} + \Sigma_{ab}\Sigma_{bb}^{-1}\mathbf{X}_b$. Conditionally on $\mathbf{X}_b = \mathbf{x}_b$, the second term is the constant $\Sigma_{ab}\Sigma_{bb}^{-1}\mathbf{x}_b$ while $\mathbf{W}$, being independent of $\mathbf{X}_b$, keeps its unconditional distribution $\mathcal{N}(\boldsymbol{\mu}_a - \Sigma_{ab}\Sigma_{bb}^{-1}\boldsymbol{\mu}_b,\; \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba})$. Adding the constant shifts the mean and gives the result. □

Two features of the formula deserve note.

- The conditional mean is *affine* in $\mathbf{x}_b$. In the bivariate case with variances σ_1^2, σ_2^2 and correlation ρ, it reads
$$\mathbb{E}(X_1 \mid X_2 = x_2) = \mu_1 + \rho\,\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2),$$
a straight line of slope $\rho\,\sigma_1/\sigma_2$, and this is exactly the *regression line* you met in last year's course on linear models. What that course posited as a modeling choice, the Gaussian world derives: *if the joint distribution is Gaussian, the best prediction of X_1 from X_2 (in mean squared error) is linear*, automatically.

- The conditional covariance $\Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$ does not depend on the observed value $\mathbf{x}_b$, and it is smaller (in the matrix order) than Σ_{aa}: observing $\mathbf{X}_b$ can only reduce uncertainty about $\mathbf{X}_a$.

Where is this theorem used? Its most direct application is *missing data*. Suppose one candidate's physics score was lost while the mathematics score is known: with a Gaussian model of the pair of scores, the theorem hands the entire distribution of the missing value given the observed one. The conditional mean is the natural imputation, and the conditional variance quantifies exactly how much the imputation can be trusted. The same computation is the workhorse of Bayesian statistics with Gaussian priors (the posterior of a Gaussian parameter given Gaussian data *is* such a conditional) and of time-series filtering, where each new observation updates the state by precisely this formula.

1.3 Affine transformations and whitening

Gaussian vectors form a family stable under affine maps: this closure property is used throughout this course.

Property 1.3.1

If $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, $A \in \mathbb{R}^{n\times d}$ and $\mathbf{b} \in \mathbb{R}^n$, then
$$A\mathbf{X} + \mathbf{b} \sim \mathcal{N}\Big(A\boldsymbol{\mu} + \mathbf{b},\ A\Sigma A^\top\Big). \tag{8}$$

Proof. We compute the characteristic function of $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$ and recognize a Gaussian one. For $\mathbf{t} \in \mathbb{R}^n$,
$$\phi_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E}\left[e^{i\mathbf{t}^\top(A\mathbf{X}+\mathbf{b})}\right] = e^{i\mathbf{t}^\top\mathbf{b}}\,\mathbb{E}\left[e^{i(A^\top\mathbf{t})^\top\mathbf{X}}\right] = e^{i\mathbf{t}^\top\mathbf{b}}\,\phi_{\mathbf{X}}(A^\top\mathbf{t}).$$
Property 1.2.1 evaluated at $A^\top\mathbf{t}$ gives
$$\phi_{\mathbf{X}}(A^\top\mathbf{t}) = \exp\left(i\,\mathbf{t}^\top A\boldsymbol{\mu} - \tfrac{1}{2}\,\mathbf{t}^\top A\Sigma A^\top\mathbf{t}\right),$$
so that
$$\phi_{\mathbf{Y}}(\mathbf{t}) = \exp\left(i\,\mathbf{t}^\top(A\boldsymbol{\mu} + \mathbf{b}) - \tfrac{1}{2}\,\mathbf{t}^\top\left(A\Sigma A^\top\right)\mathbf{t}\right),$$
which is the characteristic function of $\mathcal{N}(A\boldsymbol{\mu} + \mathbf{b},\ A\Sigma A^\top)$. Since the characteristic function determines the distribution (appendix B), the claim follows. □

Note that $A\mathbf{X} + \mathbf{b}$ is a Gaussian *vector* even when $A\Sigma A^\top$ is singular; this is another reason not to build the theory on densities.

Whitening. In one dimension there is a standard way to put a Gaussian variable on a universal scale: if $X \sim \mathcal{N}(\mu, \sigma^2)$, its *z-score*

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

(center, then divide by the spread) has one and the same distribution whatever μ and σ; this is the standardization your first-year tables of the Gaussian are built on. How should the z-score generalize to a Gaussian *vector*? Centering is unambiguous ($\mathbf{X} - \boldsymbol{\mu}$), but there is no single number σ to divide by: the spread of a vector is the whole matrix Σ. Writing the 1D division as multiplication by $(\sigma^2)^{-1/2}$ names the right generalization: multiply by $\Sigma^{-1/2}$. Take Σ invertible and let $\Sigma^{-1/2}$ be the inverse of the matrix square root $\Sigma^{1/2} = P\Lambda^{1/2}P^\top$ (appendix A). Choosing $A = \Sigma^{-1/2}$ and $\mathbf{b} = -\Sigma^{-1/2}\boldsymbol{\mu}$ in property 1.3.1 gives

$$\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}(0, \mathrm{I}_d) :$$

the multivariate z-score. Any non-degenerate Gaussian vector can be turned into the standard one by centering, rotating, rescaling and rotating back, since $\Sigma^{-1/2} = P\Lambda^{-1/2}P^\top$; the last rotation P can be skipped, as it maps $\mathcal{N}(0, \mathrm{I}_d)$ to itself. Figure 1.4 shows the first steps on data: the rotation $P^\top(\mathbf{X} - \boldsymbol{\mu})$ aligns the cloud with the axes (decorrelation), the rescaling by $\Lambda^{-1/2}$ makes it isotropic (whitening). Conversely, $\mathbf{X} = \boldsymbol{\mu} + \Sigma^{1/2}\mathbf{Z}$ with $\mathbf{Z} \sim \mathcal{N}(0, \mathrm{I}_d)$: every Gaussian vector is a linear deformation of i.i.d. standard coordinates. This is both the standard simulation recipe (the generation purpose of the prologue: to simulate synthetic rows, draw standard normals and color them) and the promised justification that the general density is the product case seen through a linear map.

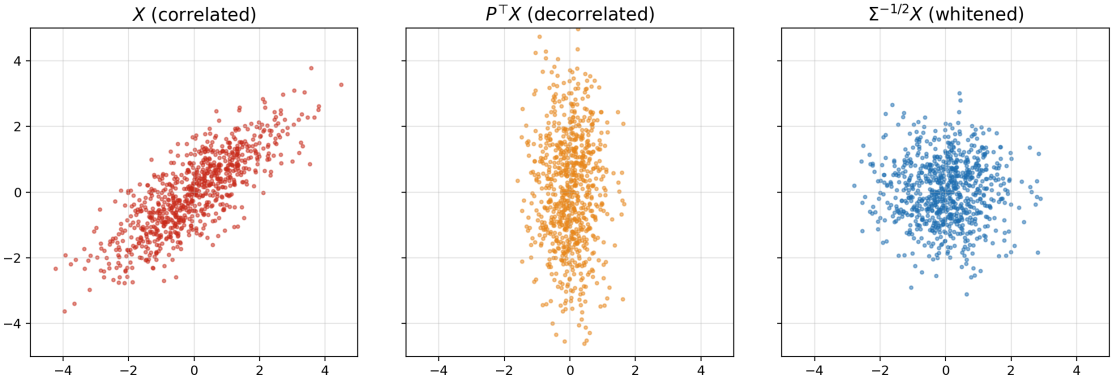


Figure 1.4: Whitening step by step: original, decorrelated ($P^\top(\mathbf{X} - \boldsymbol{\mu})$), and whitened ($\Lambda^{-1/2}P^\top(\mathbf{X} - \boldsymbol{\mu})$, which equals $\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ up to the rotation P) samples.

Whitening is also a common preprocessing step: several algorithms assume decorrelated inputs of comparable scales; independent component analysis requires it, and gradient-based optimizers converge faster on whitened features.

1.4 The Mahalanobis distance

We can now answer the anomaly question the prologue raised: is a given observation exceptional, and by how much?

Which distance should we use? Consider the entrance exam of a selective program: two papers, mathematics and physics, each marked out of 40, one row per candidate. The two scores move together: strong candidates score high on both papers, weaker ones low on both. Consider two candidates, both at Euclidean distance 10 from the mean $(18, 18)$: candidate A near $(25, 25)$,

along the correlation direction (strong across the board), and candidate B near $(25, 11)$, against it: excellent in mathematics, poor in physics. The Euclidean distance rates the two candidates equally far from the average one; the data do not (figure 1.5): A lies inside the point cloud while B is far outside it. The Euclidean distance is the wrong measure here, and the Gaussian density suggests the right one. Model ordinary candidates as $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$: the density of section 1.1 is constant exactly on the ellipsoids

$$(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = r^2,$$

so two candidates are *equally plausible* under the model precisely when they share the value of this quadratic form. Candidate A lies on a small ellipsoid (high density), candidate B on a much larger one (density smaller by orders of magnitude): if "exceptional" is to mean "implausible for an ordinary candidate", the quantity to read is not $\|\mathbf{x} - \boldsymbol{\mu}\|$ but the index r of the level set through $\mathbf{x}$. This is the definition:

Definition 1.4.1 (Mahalanobis distance)

For Σ positive definite, the *Mahalanobis distance* [1] from $\mathbf{x}$ to $\boldsymbol{\mu}$ is

$$d_\Sigma(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})}. \tag{9}$$

It measures separation in "covariance units": deviations along directions of large variance are cheap, deviations against the correlation structure are expensive, which is exactly why candidate A scores $d_\Sigma \approx 1.3$ and candidate B scores $d_\Sigma \approx 5.8$ on the same Euclidean displacement. For $\Sigma = \mathrm{I}_d$ it reduces to the Euclidean distance: the Mahalanobis distance is the Euclidean one, corrected for the shape of the cloud.

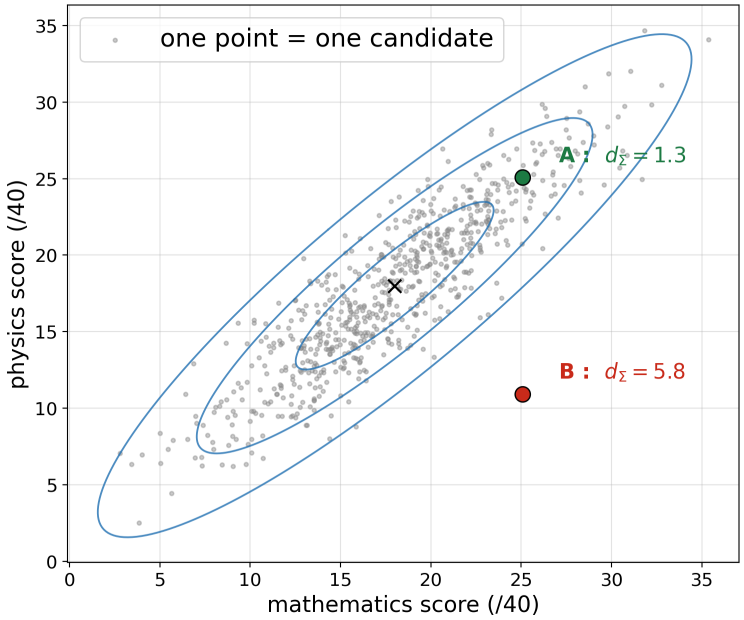


Figure 1.5: Two candidates at equal Euclidean distance from the mean candidate $\boldsymbol{\mu}$: the covariance structure makes A (strong in both papers) typical, $d_\Sigma \approx 1.3$, and B (excellent in mathematics, poor in physics) an extreme anomaly, $d_\Sigma \approx 5.8$.

Its distribution. What makes this distance a *statistical* tool rather than a mere score is that, under the model, its distribution is known exactly.

Reminder

If $Z_1, \dots, Z_p$ are i.i.d. $\mathcal{N}(0,1)$, the distribution of $\sum_{i=1}^{p} Z_i^2$ is called chi-squared with p degrees of freedom, written $\chi^2(p)$.

Proposition 1.4.1

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with Σ invertible. Then

$$d_\Sigma(\mathbf{X}, \boldsymbol{\mu})^2 = (\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu}) \;\sim\; \chi^2(d). \tag{10}$$

Proof. Whiten as in section 1.3: the vector $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ is Gaussian with mean 0 and covariance $\Sigma^{-1/2}\,\Sigma\,\Sigma^{-1/2} = \mathrm{I}_d$, i.e. $\mathbf{Z} \sim \mathcal{N}(0, \mathrm{I}_d)$: its d coordinates are i.i.d. $\mathcal{N}(0,1)$. Moreover, writing $\Sigma^{-1} = \Sigma^{-1/2}\Sigma^{-1/2}$,

$$(\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu}) = \left(\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})\right)^\top \left(\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})\right) = \|\mathbf{Z}\|^2 = \sum_{i=1}^{d} Z_i^2,$$

a sum of d squared i.i.d. standard Gaussians: a $\chi^2(d)$ by definition. □

Proposition 1.4.1 means Mahalanobis distances come *pre-calibrated*: quantiles of $\chi^2(d)$ convert a score into a probability. In dimension $d = 2$ the tail is even explicit, $\mathbb{P}(d_\Sigma^2 > r^2) = e^{-r^2/2}$, so candidate A's score is entirely unremarkable ($\mathbb{P}(d_\Sigma > 1.3) \approx 0.43$) while candidate B's is essentially impossible among ordinary candidates ($\mathbb{P}(d_\Sigma > 5.8) \approx 5 \cdot 10^{-8}$); such a copy calls for a second look: a marking error, an irregularity, or a genuinely exceptional profile. Thresholding a Mahalanobis distance at a chi-squared quantile is the basic recipe of multivariate anomaly detection, with the false-alarm rate chosen in advance.

A new scalar product, a new geometry. The Mahalanobis distance is not an isolated trick: it is the distance of a full Euclidean-style geometry that Σ installs on $\mathbb{R}^d$. Define

$$\langle \mathbf{u}, \mathbf{v} \rangle_\Sigma = \mathbf{u}^\top \Sigma^{-1} \mathbf{v}.$$

This is a genuine scalar product (bilinear and symmetric because Σ^{-1} is symmetric, positive definite because Σ^{-1} is), and its norm $\|\mathbf{u}\|_\Sigma = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle_\Sigma} = d_\Sigma(\mathbf{u}, 0)$ is exactly the Mahalanobis distance to the origin. All Euclidean vocabulary transfers (lengths, angles, orthogonality), but computed *anisotropically*. Figure 1.6 shows the two geometries side by side. In the Σ-geometry the unit ball is the covariance ellipse, and the natural coordinate grid is the image of the square grid under $\Sigma^{1/2}$, an oblique lattice whose lines are not aligned with the eigenvectors of Σ: one "unit step" is long in the direction where the data spread widely and short across it. The two red and green arrows are unit vectors of the new geometry: of unequal Euclidean length, but of unit Σ-length. The whitening map $\Sigma^{-1/2}$ of section 1.3 is precisely the change of coordinates that turns the right panel back into the left one: an *isometry* between the Σ-geometry and the standard Euclidean one, which is the geometric content of proposition 1.4.1. Several later constructions are ordinary Euclidean procedures carried out in this geometry.

1.5 Why the Gaussian is everywhere: the multivariate CLT

Everything so far assumed Gaussian data. The justification for this emphasis is that averages of variables from any distribution with finite variance become Gaussian once centered and rescaled by $\sqrt{n}$.

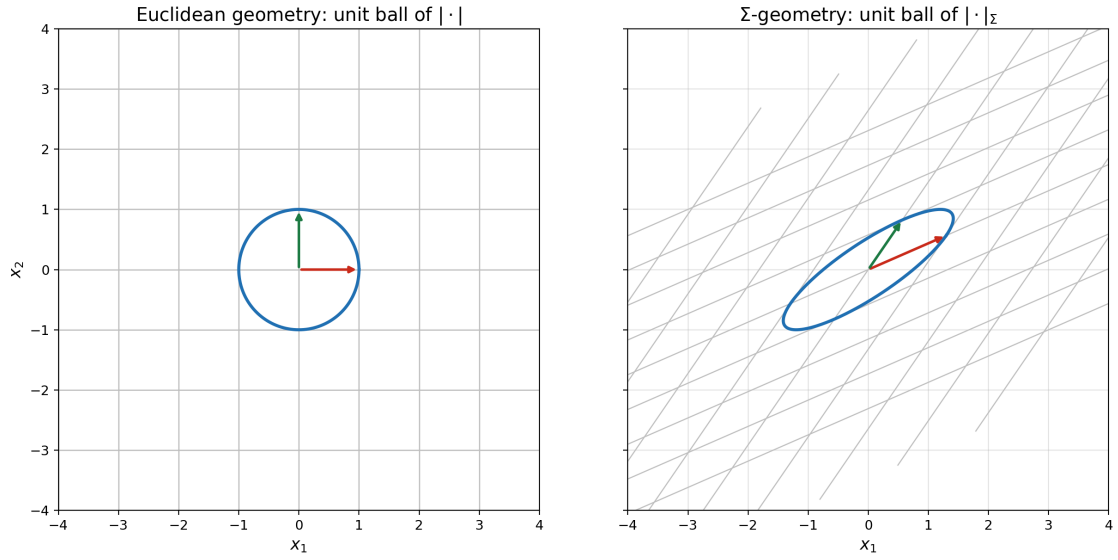


Figure 1.6: The scalar product $\langle \mathbf{u}, \mathbf{v} \rangle_\Sigma = \mathbf{u}^\top \Sigma^{-1} \mathbf{v}$ replaces the Euclidean geometry (left: square grid, unit circle) by an anisotropic one (right: the square grid mapped by $\Sigma^{1/2}$, an oblique lattice; the unit ball is the covariance ellipse). The arrows are the images of the standard basis: unit vectors of the new geometry.

Theorem 1.5.1 (Multivariate central limit theorem)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be i.i.d. random vectors from any distribution with mean $\boldsymbol{\mu}$ and finite covariance Σ, and let $\bar{\mathbf{X}}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i$. Then

$$\sqrt{n}\,(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow[n\to\infty]{d} \mathcal{N}(0, \Sigma). \tag{11}$$

The proof is a textbook application of the Cramér–Wold device (appendix B): for any fixed direction $\mathbf{a} \in \mathbb{R}^d$, the scalars $\mathbf{a}^\top \mathbf{X}_i$ are i.i.d. with mean $\mathbf{a}^\top \boldsymbol{\mu}$ and variance $\mathbf{a}^\top \Sigma\, \mathbf{a}$, so the one-dimensional CLT gives $\sqrt{n}\, \mathbf{a}^\top (\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(0, \mathbf{a}^\top \Sigma\, \mathbf{a})$; since this holds for every $\mathbf{a}$, the device gives the vector limit.

Note what the theorem does *not* assume: no density, no symmetry, no Gaussianity of the data; it needs only i.i.d. sampling and finite second moments. And note what it concludes: the limiting covariance is the covariance Σ *of the data themselves*, so the covariances, hence the correlations, of the coordinates survive in the limit; any nonlinear dependence between them disappears. Figure 1.7 illustrates both points on a deliberately skewed 2D distribution with correlated coordinates (correlation about 0.63): at $n = 1$ the cloud is asymmetric and fan-shaped; by $n = 200$ the rescaled means form a clean elliptical Gaussian cloud whose tilt reflects the correlation of the original data. This theorem justifies Gaussian modeling for aggregates: when the quantity of interest is an average, a sum, or a net effect of many contributions, the Gaussian family is asymptotically correct.

1.6 Estimating a Gaussian model

1.6.1 Maximum likelihood estimator

It remains to fit the model. Everything in this section is the univariate estimation story you rehearsed in first year, with vectors and matrices substituted for scalars, plus one new phenomenon at the end that has no univariate counterpart.

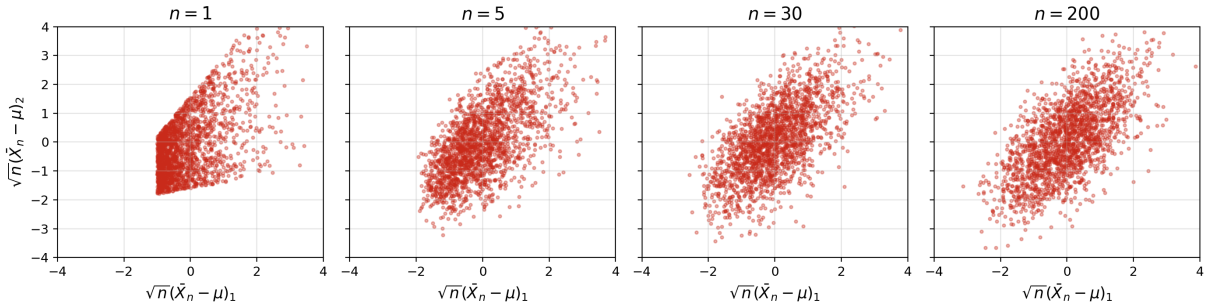


Figure 1.7: Rescaled means $\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ of a skewed 2D distribution with correlated coordinates: an elliptical Gaussian cloud, tilted by that correlation, emerges as n grows.

Reminder

(*Likelihood and MLE in dimension 1.*) The *likelihood* of an i.i.d. sample is its joint density read as a function of the parameters, $L(\theta) = \prod_{i=1}^n f_\theta(x_i)$: it grades each candidate θ by how unsurprising it makes the observed data, and the *maximum likelihood estimator* (MLE) picks the best-graded one. For $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$: $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_i (x_i - \bar{x})^2$; the latter is biased, $\mathbb{E}(\hat{\sigma}^2_{\text{MLE}}) = \frac{n-1}{n}\sigma^2$, whence the unbiased $\hat{\sigma}^2_u$ dividing by $n-1$. The estimators fluctuate with exactly known laws: $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$, $(n-1)\,\hat{\sigma}^2_u/\sigma^2 \sim \chi^2(n-1)$, and $\bar{X} \perp\!\!\!\perp \hat{\sigma}^2_u$ (Cochran's theorem).

The likelihood. Let $\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with Σ invertible. By independence, the likelihood of the observed sample is

$$L(\boldsymbol{\mu}, \Sigma) = \prod_{i=1}^n \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\tfrac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu})\right). \tag{12}$$

Theorem 1.6.1 (Gaussian MLE)

If the matrix $\widehat{\Sigma}_{\text{MLE}}$ below is invertible (which requires $n \geq d+1$), the likelihood is maximized at

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i, \qquad \widehat{\Sigma}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top. \tag{13}$$

If $\widehat{\Sigma}_{\text{MLE}}$ is singular, the likelihood is unbounded and has no maximizer.

The estimators have exactly the shape the reminder announces: empirical mean, empirical average of (outer-product) squared deviations. The proof is a beautiful piece of matrix calculus: the mean part works as in 1D, while the covariance part hinges on rewriting the exponent with the trace trick $\sum_i (\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) = \operatorname{tr}(\Sigma^{-1} S_{\boldsymbol{\mu}})$, where $S_{\boldsymbol{\mu}} = \sum_i (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^\top$, and optimizing over the eigenvalues of $\Sigma^{-1/2}$-conjugated matrices.

Fluctuations. As in 1D, the estimators are random and their laws are exactly known. The law of a random *matrix* such as $\widehat{\Sigma}_{\text{MLE}}$ requires one new named distribution, which plays for covariance matrices the role the chi-squared plays for variances:

Definition 1.6.1 (Wishart distribution)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n \sim \mathcal{N}(0, \Sigma)$ be i.i.d. The distribution of the random positive semi-definite matrix $\sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^\top$ is called the *Wishart distribution* $\mathcal{W}_d(n, \Sigma)$.

For $d = 1$ and $\Sigma = 1$ the definition reads $\sum_i X_i^2 \sim \chi^2(n)$: the Wishart *is* the multivariate chi-squared. The connection persists in any dimension through one-dimensional projections: for any direction $\mathbf{a}$ with $\mathbf{a}^\top \Sigma\, \mathbf{a} \neq 0$, and $V \sim \mathcal{W}_d(n, \Sigma)$,

$$\frac{\mathbf{a}^\top V \mathbf{a}}{\mathbf{a}^\top \Sigma\, \mathbf{a}} \sim \chi^2(n),$$

since $\mathbf{a}^\top V \mathbf{a} = \sum_i (\mathbf{a}^\top \mathbf{X}_i)^2$ is a sum of n squared i.i.d. Gaussians of variance $\mathbf{a}^\top \Sigma\, \mathbf{a}$.

Property 1.6.1

Under the model above, with $\widehat{\Sigma}_u = \frac{1}{n-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$ the unbiased covariance estimator:

$$\bar{\mathbf{X}} \sim \mathcal{N}\Big(\boldsymbol{\mu}, \tfrac{\Sigma}{n}\Big), \qquad (n-1)\, \widehat{\Sigma}_u \sim \mathcal{W}_d(n-1, \Sigma), \qquad \bar{\mathbf{X}} \perp\!\!\!\perp \widehat{\Sigma}_u. \tag{14}$$

Moreover $\mathbb{E}(\widehat{\Sigma}_{\text{MLE}}) = \frac{n-1}{n}\Sigma$, i.e. $\widehat{\Sigma}_u = \frac{n}{n-1}\, \widehat{\Sigma}_{\text{MLE}}$ is the unbiased version: the MLE covariance is biased, for the same degree-of-freedom reason as in 1D.

The first claim is immediate from the affine stability of section 1.3 ($\bar{\mathbf{X}}$ is a linear map of the Gaussian vector $(\mathbf{X}_1, \dots, \mathbf{X}_n)$); the Wishart law and the independence are the multivariate Cochran theorem, which we state without proof. The table below collects the univariate–multivariate correspondence built over the last five sections:

	dimension 1 (first year)	dimension d (this chapter)
model	$\mathcal{N}(\mu, \sigma^2)$	$\mathcal{N}(\boldsymbol{\mu}, \Sigma)$
parameters	2	$d + d(d+1)/2$
MLE of the mean	$\bar{x}$	$\bar{\mathbf{x}} \in \mathbb{R}^d$
MLE of the (co)variance	$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_i (x_i - \bar{x})^2$	$\widehat{\Sigma}_{\text{MLE}} = \frac{1}{n}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$
unbiased estimator	$\hat{\sigma}^2_u$ $(\div\, n{-}1)$	$\widehat{\Sigma}_u$ $(\div\, n{-}1)$
law of the mean	$\mathcal{N}(\mu, \sigma^2/n)$	$\mathcal{N}(\boldsymbol{\mu}, \Sigma/n)$
law of the (co)variance	$(n{-}1)\hat{\sigma}^2_u/\sigma^2 \sim \chi^2(n{-}1)$	$(n{-}1)\widehat{\Sigma}_u \sim \mathcal{W}_d(n{-}1, \Sigma)$
standardized deviation	$(x-\mu)^2/\sigma^2$	$d_\Sigma(\mathbf{x}, \boldsymbol{\mu})^2 \sim \chi^2(d)$

1.6.2 The curse of dimension

Consider the parameters row of the table. In one dimension the Gaussian model asks for two numbers. In dimension d it asks for d means plus one covariance entry per pair of coordinates, the d variances included:

$$k \;=\; d + \frac{d(d+1)}{2}, \qquad \begin{array}{c|cccc} d & 2 & 10 & 100 & 1000 \\ \hline k & 5 & 65 & 5\,150 & 501\,500 \end{array} \tag{15}$$

The mean vector grows linearly; the covariance grows *quadratically*, and it dominates the parameter count from the start. This is the *curse of dimension* in its statistical guise. Your first-year course on dimension reduction showed you its geometric one (volumes and distances misbehaving as d grows); here it is estimation accuracy that suffers. Estimating Σ well is the hard part of fitting a Gaussian model, and the difficulty comes in two regimes.

If $n \leq d$, the estimate is singular, always. This is not bad luck on a particular dataset but a theorem:

Proposition 1.6.1

$\operatorname{rank}(\widehat{\Sigma}_{\text{MLE}}) \leq \min(d,\, n-1)$. In particular, whenever $n \leq d$, the sample covariance $\widehat{\Sigma}_{\text{MLE}}$ is singular.

Proof. Write $\widehat{\Sigma}_{\text{MLE}} = \frac{1}{n}\mathbf{C}^\top\mathbf{C}$, where $\mathbf{C} \in \mathbb{R}^{n\times d}$ stacks the centered rows $(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. A matrix and its Gram matrix have the same rank, so $\operatorname{rank}(\widehat{\Sigma}_{\text{MLE}}) = \operatorname{rank}(\mathbf{C})$, and $\operatorname{rank}(\mathbf{C}) \le \min(n, d)$ trivially. One more unit is lost to centering: the rows of $\mathbf{C}$ sum to $\sum_i(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$, so any one row is determined by the $n-1$ others, and $\operatorname{rank}(\mathbf{C}) \le n-1$. Hence $\operatorname{rank}(\widehat{\Sigma}_{\text{MLE}}) \le \min(d, n-1) < d$ whenever $n \le d$. □

A singular $\widehat{\Sigma}_{\text{MLE}}$ means the fitted Gaussian is *degenerate* in the sense of section 1.1: it concentrates on the affine subspace through $\bar{\mathbf{x}}$ spanned by the centered rows $\mathbf{x}_i - \bar{\mathbf{x}}$ (of dimension at most $n-1$), has no density, and $\widehat{\Sigma}^{-1}_{\text{MLE}}$ does not exist, so the Mahalanobis distance, the anomaly detector and the conditional formulas of theorem 1.2.2 (which need Σ_{bb}^{-1}) are simply unavailable. Nor is $n \le d$ an exotic regime: a genomics dataset routinely records $d = 20\,000$ genes for a few hundred samples.

If $n > d$, the estimate is invertible, but its accuracy degrades as d grows. Figure 1.8 shows a simulation with the simplest possible truth: data from $\mathcal{N}(0, \mathrm{I}_d)$, whose true eigenvalues all equal 1, tracking for each dimension the *largest* and *smallest* eigenvalue of $\widehat{\Sigma}_{\text{MLE}}$ as the sample grows. At $d = 2$, both are close to the truth already for modest n. At $d = 50$, a sample of $n = 100$ (twice d) still returns eigenvalues from about 0.1 to 2.8: the estimator overstates the anisotropy of the data, inflating the top of the spectrum and deflating the bottom, although all true eigenvalues are equal. At $d = 200$, thousands of observations are needed before the spectrum tightens. The reason is a counting argument: the sample provides $n \times d$ numbers and the model asks for $d(d+1)/2$ parameters, so accuracy is governed not by n alone but by the ratio n/d. The practical consequence is that the model's most useful formulas all use $\widehat{\Sigma}^{-1}_{\text{MLE}}$, whose largest eigenvalue $1/\lambda_{\min}(\widehat{\Sigma}_{\text{MLE}})$ comes from the eigenvalue the estimator underestimates most: inverting amplifies exactly the worst error.

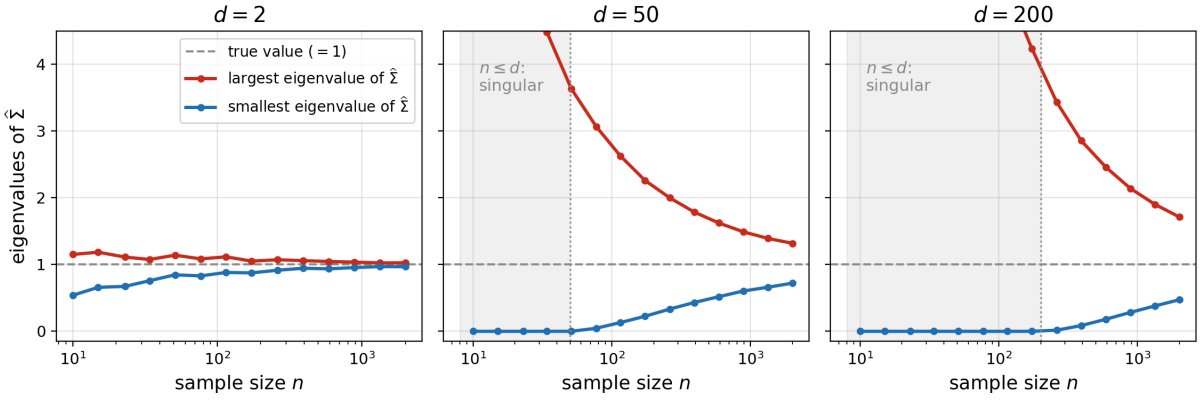


Figure 1.8: Largest and smallest eigenvalues of $\widehat{\Sigma}_{\text{MLE}}$ (averaged over 20 replications) as the sample size grows, on data from $\mathcal{N}(0, \mathrm{I}_d)$ where every true eigenvalue is 1. Left to right: $d = 2$, 50, 200. In the shaded region $n \le d$ the smallest eigenvalue is exactly 0 (proposition 1.6.1); beyond it, convergence toward the truth is slower the larger d.

1.6.3 Regularization

The difficulty just diagnosed is, at bottom, an *inference* problem: we want to infer $d + d(d+1)/2$ parameters and we only have n observations to do it. The obvious solution is to get more data, and it is the right one when available: in figure 1.8, every curve does converge to the truth as n grows. But in practice more data is often not an option: the sample is whatever the measurement campaign, the archive, or the budget produced. Two approaches remain, and both are called regularization. Either we *reduce the parameter count*, restricting the family until n suffices for it

(regularizing the *problem*), or we keep the family and *accept a bias in the inference* in exchange for stability (regularizing the *estimator*).

Regularize the problem: fewer parameters, more assumptions. Nothing forces the family to contain every covariance matrix. Assume Σ *diagonal* and the count collapses from $d + d(d+1)/2$ to $2d$; assume it *spherical*, $\Sigma = \sigma^2 \mathrm{I}_d$, and it collapses to $d+1$. These are modeling assumptions in the full sense of the chapter's opening reminder: the diagonal family asserts that the coordinates are uncorrelated. For the exam table of section 1.4, it asserts that knowing a candidate's mathematics score says nothing about their physics score, which is exactly the structure the Mahalanobis section exploited. The gain is estimability at small n; the cost is that the model can no longer represent dependence at all. Whether this is acceptable depends on the question asked.

Regularize the estimator: same model, biased but stable. The alternative keeps the full family (the parameter count stands) but replaces $\widehat{\Sigma}_{\mathrm{MLE}}$ by a blend with a fixed, well-behaved target, in its simplest form

$$\widehat{\Sigma}_\alpha \;=\; (1-\alpha)\,\widehat{\Sigma}_{\mathrm{MLE}} \;+\; \alpha\,\tau \mathrm{I}_d, \qquad \alpha \in (0,1], \tag{16}$$

with $\tau > 0$ a scale (say the average sample variance). The name *regularization* is meant literally. The defect of $\widehat{\Sigma}_{\mathrm{MLE}}$ in high dimension is its *irregularity*: it is singular or nearly so, hence not invertible, or invertible with an inverse that blows noise up without bound. The blend repairs this: since eigenvalues of a sum of symmetric matrices are bounded below by the sums of the smallest ones,

$$\lambda_{\min}(\widehat{\Sigma}_\alpha) \;\geq\; (1-\alpha)\,\underbrace{\lambda_{\min}(\widehat{\Sigma}_{\mathrm{MLE}})}_{\geq 0} + \alpha\tau \;\geq\; \alpha\tau \;>\; 0\,:$$

$\widehat{\Sigma}_\alpha$ is invertible *by construction*, for any n and d, and its inverse is stable. Pulling the estimate toward a fixed target thus turns an ill-posed estimation problem into a well-posed one. The price is a bias: $\widehat{\Sigma}_\alpha$ is pulled toward $\tau \mathrm{I}_d$ even if the truth is far from it, and the weight α arbitrates between the variance of $\widehat{\Sigma}_{\mathrm{MLE}}$ and the bias of the target. Choosing α from the data can itself be automated (the Ledoit–Wolf estimator is the standard reference). How such trade-offs are tuned in general recurs throughout the course.

1.7 Conclusion

Figure 1.9 shows the same modeling act carried out in dimensions one, two, and three: a cloud of points and a fitted density. In every dimension the fit is pinned down by two objects only, a mean and a covariance. The Gaussian is the simplest multivariate distribution characterized by these two objects, and what it represents is correspondingly simple: a *single cluster* of data, symmetric around its center, whose shape is well approximated by an ellipsoid, the covariance defining the geometry of that ellipsoid, its orientation and its elongations. Everything the chapter did unfolds from this picture: the density's level sets *are* the ellipsoids; the Mahalanobis distance measures how many ellipsoids away a point sits, calibrated by the chi-squared law; whitening is the change of variables that makes the ellipsoid a sphere; estimation fills in the two objects from data, at a price, quadratic in d, that the covariance dictates.

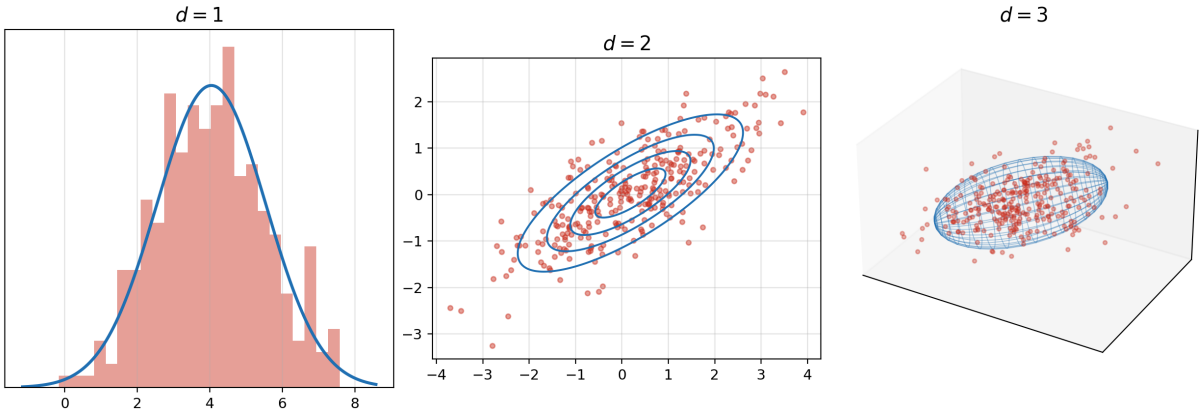


Figure 1.9: One Gaussian cluster in $d = 1, 2, 3$: data (red) and the fitted Gaussian (blue). In every dimension the fitted density is characterized by its mean and covariance alone: mass symmetric around the mean, level sets ellipsoids whose orientation and elongations the covariance defines.

One center, one ellipsoid: that is also the model's limit. Many datasets gather around *several* centers at once (customers of two kinds, cells of two subtypes), and no single ellipsoid can be symmetric around all of them. Describing such data with several Gaussians simultaneously, and fitting them when no column says which point belongs to which, is the subject of the next chapter.

A Linear algebra toolbox

Spectral theorem. Every symmetric matrix $\mathbf{A} \in \mathbb{S}_d$ admits d real eigenvalues $\lambda_1, \dots, \lambda_d$ with orthonormal eigenvectors $\mathbf{e}_1, \dots, \mathbf{e}_d$: $\mathbf{A}\mathbf{e}_i = \lambda_i \mathbf{e}_i$. In matrix form $\mathbf{A} = P\Lambda P^\top$ with $P = (\mathbf{e}_1 | \cdots | \mathbf{e}_d)$ orthogonal ($PP^\top = \mathrm{I}_d$) and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$. Equivalently, $\mathbf{A} = \sum_i \lambda_i \mathbf{e}_i \mathbf{e}_i^\top$: a weighted sum of rank-one projectors.

Positive matrices. $\mathbf{A}$ is positive semi-definite ($\mathbf{A} \succeq 0$, all $\lambda_i \geq 0$) iff its quadratic form $Q(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}$ is nonnegative for all $\mathbf{x}$; positive definite when strict. Covariance matrices are positive semi-definite since $\mathbf{a}^\top \Sigma\, \mathbf{a} = \mathbb{V}(\mathbf{a}^\top \mathbf{X}) \geq 0$. Moreover $\mathrm{tr}(\mathbf{A}) = \sum_i \lambda_i$, $\det(\mathbf{A}) = \prod_i \lambda_i$, and $\mathrm{rank}(\mathbf{A})$ is the number of nonzero eigenvalues.

Matrix square root. For $\mathbf{A} \succeq 0$, $\mathbf{A}^{1/2} = P\Lambda^{1/2}P^\top$ with $\Lambda^{1/2} = \mathrm{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_d})$; it is the unique positive semi-definite matrix whose square is $\mathbf{A}$. When $\mathbf{A} \succ 0$, $\mathbf{A}^{-1/2} = (\mathbf{A}^{1/2})^{-1} = P\Lambda^{-1/2}P^\top$.

Quadratic forms and ellipsoids. For $\mathbf{A} \succ 0$, the region $\{\mathbf{x} : (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{A}(\mathbf{x} - \mathbf{x}_0) = r^2\}$ is an ellipsoid centered at $\mathbf{x}_0$ with principal axes the eigenvectors of $\mathbf{A}$ and semi-axis lengths $r/\sqrt{\lambda_i(\mathbf{A})}$. Applied to $\mathbf{A} = \Sigma^{-1}$, whose eigenvalues are $1/\lambda_i(\Sigma)$, this gives semi-axes $r\sqrt{\lambda_i(\Sigma)}$, the geometry of section 1.1.

B Random vectors toolbox

Expectation and covariance. For a random vector $\mathbf{X} = (X_1, \dots, X_d)^\top$, the expectation is taken componentwise, $\mathbb{E}(\mathbf{X}) = (\mathbb{E}X_1, \dots, \mathbb{E}X_d)^\top$, and for any matrix A and vector $\mathbf{b}$, $\mathbb{E}(A\mathbf{X}+\mathbf{b}) = A\mathbb{E}(\mathbf{X}) + \mathbf{b}$. The covariance matrix

$$\mathbb{V}(\mathbf{X}) = \Sigma = \mathbb{E}\left[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\right] = \mathbb{E}(\mathbf{X}\mathbf{X}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$$

collects variances on the diagonal and covariances $\sigma_{X_iX_j}$ off it. Key identities: $\mathbb{V}(\mathbf{a}^\top\mathbf{X}) = \mathbf{a}^\top\Sigma\,\mathbf{a}$, $\mathbb{V}(A\mathbf{X}+\mathbf{b}) = A\Sigma A^\top$, and $\Sigma \succeq 0$ always.

Correlation. $\rho_{X_iX_j} = \sigma_{X_iX_j}/\sqrt{\mathbb{V}(X_i)\mathbb{V}(X_j)} \in [-1,1]$. Independence implies zero correlation; the converse is false in general (it holds for Gaussian vectors, section 1.2).

Characteristic function. $\phi_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}[e^{i\mathbf{t}^\top\mathbf{X}}]$, $\mathbf{t}\in\mathbb{R}^d$, determines the distribution of $\mathbf{X}$ entirely. Componentwise: $\phi_{X_1}(t_1) = \phi_{\mathbf{X}}(t_1,0,\dots,0)$. The components $X_1,\dots,X_d$ are independent iff $\phi_{\mathbf{X}}(\mathbf{t}) = \prod_j \phi_{X_j}(t_j)$ for all $\mathbf{t}$.

Cramér–Wold theorem. The distribution of $\mathbf{X}\in\mathbb{R}^d$ is entirely determined by the collection of univariate distributions of $\mathbf{t}^\top\mathbf{X}$ for all $\mathbf{t}\in\mathbb{R}^d$. It is the bridge between univariate and multivariate limit theorems, and the reason definition 1.1.1 characterizes a distribution at all. Its version for convergence, the *Cramér–Wold device*, reads

$$\mathbf{X}_n \xrightarrow{d} \mathbf{X} \iff \mathbf{a}^\top\mathbf{X}_n \xrightarrow{d} \mathbf{a}^\top\mathbf{X} \quad \text{for every } \mathbf{a}\in\mathbb{R}^d:$$

convergence in distribution of random vectors reduces to that of all their one-dimensional projections.

References

[1] P. C. Mahalanobis. On the generalized distance in statistics. *Proceedings of the National Institute of Sciences of India*, 2(1):49–55, 1936.

Chapter 2
Gaussian mixtures: complexity and model selection

Chapter 1 closed on its own limit: one Gaussian describes one cluster (one center, one ellipsoid), and many datasets gather around *several* centers at once. A marketing team suspects a few distinct "types" among thousands of unlabeled customers; a biologist looks for unknown cell subtypes in gene-expression data. Discovering groups in unlabeled data is *clustering*, our first *unsupervised* problem: no labels exist, and no "right answers", only structure to be found. This chapter builds the family that models such data, several Gaussians at once: the Gaussian mixture. Its likelihood cannot be maximized directly as in chapter 1, and the algorithm that maximizes it, EM, is one of the most used algorithms in statistics.

2.1 From one Gaussian to several groups

Two groups, one histogram. Begin in one dimension, with no units attached. Figure 2.1 shows the histogram of a sample of 400 values: two clearly separated bumps of equal size. No single Gaussian density can produce this shape (a $\mathcal{N}(\mu, \sigma^2)$ has one peak), yet each bump, taken alone, looks perfectly Gaussian. Suppose for a moment we *knew* which points belong to which bump. Then nothing new would be needed: split the sample in two and fit one Gaussian per group by chapter 1's maximum likelihood: a mean and a variance $(\hat{\mu}_1, \hat{\sigma}_1^2)$ for the left group, another pair for the right. The two fits also combine into a single model for a point whose group we ignore. The split being half-and-half, a random point is produced by a fair coin flip between the two Gaussians, and its density is the corresponding *superposition*, half of each bump:

$$f(x) = \tfrac{1}{2}\,\varphi(x;\,\mu_1, \sigma_1^2) + \tfrac{1}{2}\,\varphi(x;\,\mu_2, \sigma_2^2), \tag{17}$$

where $\varphi(\cdot\,; \mu, \sigma^2)$ denotes the $\mathcal{N}(\mu, \sigma^2)$ density. This is the solid curve of figure 2.1: a two-peaked density, assembled from two one-peaked ones.

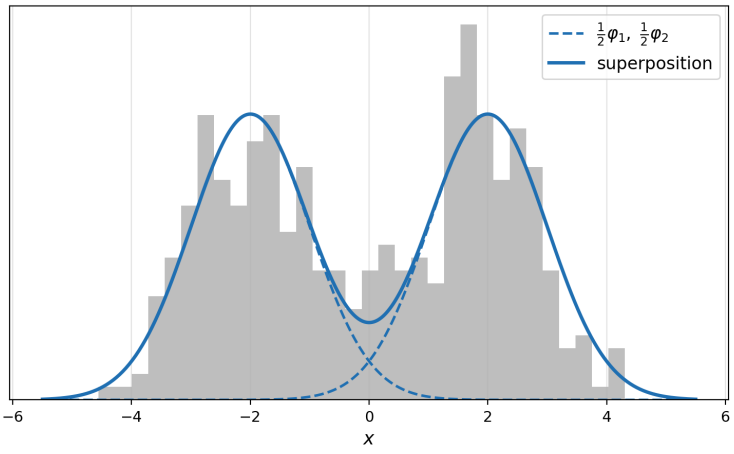


Figure 2.1: A sample with two equal groups (gray histogram), the two half-weighted Gaussian densities (dashed), and their superposition (solid): a two-peaked density that no single Gaussian can produce.

Remark

Equation (17) mixes *densities*, not random variables: it is not the distribution of the average $\frac{1}{2}(X_1 + X_2)$ of two independent Gaussian variables. That average is again Gaussian, $\mathcal{N}(\frac{\mu_1+\mu_2}{2}, \frac{\sigma_1^2+\sigma_2^2}{4})$, with a single peak between the two centers. Under the density (17), an observation is drawn from *one* of the two Gaussians, chosen at random, never from both at once.

A concrete example. Figure 2.2 shows the kind of data this machinery is for: a store's loyalty-card database: one row per customer, monthly visits against monthly spend, exactly as the file presents it. The eye needs no help: there are clearly *three* groups of customers: a large cloud of occasional shoppers in the lower left (a few visits, small baskets); a group of regulars (around ten visits a month, medium spend); and a smaller group of bulk buyers, who come less often than the regulars but spend the most of anyone. And each group, taken alone, has exactly the shape chapter 1's conclusion described: a single cluster, roughly symmetric around its center, well approximated by an ellipsoid: a Gaussian cluster.

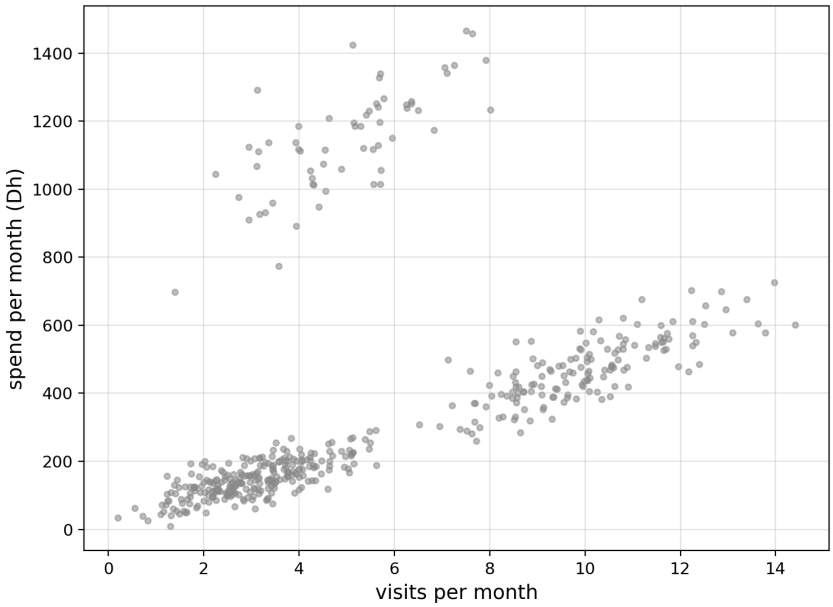


Figure 2.2: A store's loyalty-card customers: monthly visits against monthly spend, as the file presents them, with no labels. Three groups of different sizes are plainly visible, and each one, taken alone, looks like a Gaussian cluster.

Determining these Gaussians is worth real money. The three means are three *customer profiles* (the typical occasional, regular and bulk customer) around which the store can build segments, offers and stock decisions; the three covariances say how variable each segment is; the group sizes say how much of the customer base each represents. And once the groups are determined, each customer can be *assigned*: which segment does this card belong to, and with what confidence? The natural model is the superposition idea of equation (17), upgraded to what the picture demands: several Gaussians on $\mathbb{R}^2$ rather than two on $\mathbb{R}$; weights that are *not* equal, since the three groups have visibly different sizes; and, above all, *no labels*: no column of the file says which customer belongs to which group, and nobody has ever annotated them. Can the parameters of the Gaussians, the proportions, and the memberships all be recovered from the gray cloud alone? The groups remain visible without any annotation, so the information survives in the shape of the data; the rest of the chapter turns this observation into an estimator.

2.2 Latent variables and the mixture model

Hidden labels as random variables. To reason about the missing labels, give them a name: let $Z \in \{1, \dots, K\}$ be the group of an observation, a genuine random variable, with $\mathbb{P}(Z = k) = \pi_k$ and $\mathbf{X} \mid Z = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$, that the model contains but the data do not show: the variable Z is not observed in the data. Such a variable is called *latent*, and the model is a *latent variable model*. The process that generates one observation is assumed to be as follows:

1. *pick a group*: choose k with probability π_k;
2. *draw the point*: $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$;
3. *forget the group*: record $\mathbf{X}$, discard k.

Step 3 is exactly what the customer file shows: the groups exist in the process that produced the data, but not in the empirical data.

Marginalizing the label. What distribution does the recipe produce for the $\mathbf{X}$ we actually observe? The law of total probability answers this. The events $\{Z = 1\}, \dots, \{Z = K\}$ partition the sample space, so for any region $A \subset \mathbb{R}^d$,

$$\begin{aligned}\mathbb{P}(\mathbf{X} \in A) &= \sum_{k=1}^{K} \mathbb{P}(Z = k)\, \mathbb{P}(\mathbf{X} \in A \mid Z = k) \\ &= \sum_{k=1}^{K} \pi_k \int_A \varphi(\mathbf{x};\, \boldsymbol{\mu}_k, \Sigma_k)\, \mathrm{d}\mathbf{x} = \int_A \Big(\sum_{k=1}^{K} \pi_k\, \varphi(\mathbf{x};\, \boldsymbol{\mu}_k, \Sigma_k) \Big) \mathrm{d}\mathbf{x},\end{aligned}$$

where step 2 of the recipe supplies the conditional distribution $\mathbf{X} \mid Z = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$. The probability of every region is obtained by integrating one fixed function, which is therefore the marginal density of $\mathbf{X}$:

$$f(\mathbf{x}) = \sum_{k=1}^{K} \pi_k\, \varphi(\mathbf{x};\, \boldsymbol{\mu}_k, \Sigma_k). \tag{18}$$

This is a *mixture* of Gaussians, a weighted superposition of bumps. For two groups of equal weight, equation (18) recovers the half-and-half superposition (17); the general formula frees both the number of groups K and their proportions π_k, which need not be equal (the customer cloud of figure 2.2, with its three groups of visibly different sizes, needs $K = 3$ and three different weights) and need not be known.

Definition 2.2.1 (Gaussian mixture model [1])

A random vector $\mathbf{X}$ on $\mathbb{R}^d$ follows a *Gaussian mixture distribution* with K components and parameters

$$\theta = (\pi_1, \dots, \pi_K,\, \boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K,\, \Sigma_1, \dots, \Sigma_K), \qquad \pi_k > 0, \quad \textstyle\sum_k \pi_k = 1,$$

when its density is the mixture (18). The *Gaussian mixture model* assumes that the observations $\mathbf{X}_1, \dots, \mathbf{X}_n$ are i.i.d. draws from such a distribution, with θ unknown.

Identifiability. Take $K = 2$ and any parameter vector $\theta = (\pi_1, \pi_2, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma_1, \Sigma_2)$, and let θ' be the same parameters with the two components numbered the other way, $\theta' = (\pi_2, \pi_1, \boldsymbol{\mu}_2, \boldsymbol{\mu}_1, \Sigma_2, \Sigma_1)$. Substituting θ' into (18),

$$f(\mathbf{x};\, \theta') = \pi_2\, \varphi(\mathbf{x};\, \boldsymbol{\mu}_2, \Sigma_2) + \pi_1\, \varphi(\mathbf{x};\, \boldsymbol{\mu}_1, \Sigma_1) = f(\mathbf{x};\, \theta),$$

because addition is commutative: $\theta \neq \theta'$ as soon as the two components differ, yet the two parameter vectors define exactly the same distribution. The map $\theta \mapsto f(\cdot\,;\theta)$ is therefore not one-to-one (the parameters are said to be *non-identifiable*), and the same argument works for any permutation of the K components. The practical consequence is that the *indexing* of the components is not reliable: nothing in the data makes one of them "component 1", and two runs of an estimation procedure may return the same fitted mixture with the labels swapped. Section 2.8 examines what this does and does not break.

Membership probabilities. The model can also answer the assignment question of section 2.1: which group produced a given observation? Elementary conditional probability is enough. The joint weight of the event "group k *and* a point at $\mathbf{x}$" is $\pi_k\,\varphi(\mathbf{x};\boldsymbol{\mu}_k,\Sigma_k)$ (the probability of picking the group times the density it puts at $\mathbf{x}$), and the total weight of "a point at $\mathbf{x}$", whatever the group, is their sum: the mixture density $f(\mathbf{x})$. Dividing the one by the other is Bayes' rule, written first with its ingredients named and then expanded using the model:

$$\gamma_k(\mathbf{x}) = \mathbb{P}(Z=k \mid \mathbf{X}=\mathbf{x}) = \frac{\mathbb{P}(Z=k)\; f(\mathbf{x} \mid Z=k)}{f(\mathbf{x})} = \frac{\pi_k\,\varphi(\mathbf{x};\,\boldsymbol{\mu}_k,\Sigma_k)}{\sum_{j=1}^{K}\pi_j\,\varphi(\mathbf{x};\,\boldsymbol{\mu}_j,\Sigma_j)}, \tag{19}$$

the denominator being the mixture density (18). These posterior probabilities are called *responsibilities*: a *soft* group assignment summing to one. A customer deep inside the occasional cloud has responsibility near 1 for that component; a customer sitting between two groups comes out near one half for each: the soft assignment records the ambiguity instead of hiding it. Clustering is now an inference problem: estimate θ, then read each customer's group off its responsibilities.

2.3 The EM algorithm

A chicken-and-egg problem. Section 2.1 left us with two half-problems, each easy *on its own*. If the labels were known (as we briefly pretended for the two-bump sample), estimating the parameters would be chapter 1's MLE applied group by group. If the parameters were known, computing each point's membership probabilities would be the responsibilities formula of section 2.2. Each half is easy given the other, and neither is available. The EM algorithm [2] breaks this circle by *alternating*: start from a guess of the parameters, compute the memberships they imply, re-estimate the parameters treating those soft memberships as if they were data, and repeat.

Why direct maximization fails. The log-likelihood of a mixture,

$$\ell(\theta) = \sum_{i=1}^{n} \log\left(\sum_{k=1}^{K}\pi_k\,\varphi(\mathbf{x}_i;\,\boldsymbol{\mu}_k,\Sigma_k)\right), \tag{20}$$

has a *logarithm of a sum* where chapter 1's Gaussian log-likelihood had a sum of logarithms. Each of those logarithms was an explicit function of $(\boldsymbol{\mu},\Sigma)$, which is what let chapter 1 maximize the likelihood in closed form; with the sum trapped inside the logarithm this simplification no longer occurs, and setting gradients to zero yields coupled equations with no closed form.

Component indicators and the complete-data likelihood. To make the chicken-and-egg idea exact, encode the hidden label of observation i as *indicator variables*

$$z_{ik} = \mathbf{1}\{Z_i = k\} \qquad (z_{ik}\in\{0,1\},\ \textstyle\sum_k z_{ik}=1)\,:$$

the one-hot vector $(z_{i1},\dots,z_{iK})$ says which component produced point i. Suppose the indicators *were* observed alongside the data: the sample would then consist of the pairs $(\mathbf{x}_i, z_i)$, and its

likelihood, the *complete-data likelihood* L_c, is built from the *joint* density of $(\mathbf{X}, Z)$, evaluated at those pairs. This is a different object from the observed-data likelihood (20), which uses only the marginal density of $\mathbf{X}$. The joint density of one pair follows from the generative recipe of section 2.2 (pick the group, then draw the point), so for a pair with label $z_i = k$,

$$p(\mathbf{x}_i, z_i;\, \theta) = \mathbb{P}(Z_i = k)\; f(\mathbf{x}_i \mid Z_i = k) = \pi_k\, \varphi(\mathbf{x}_i;\, \boldsymbol{\mu}_k, \Sigma_k).$$

The indicators write this without case distinctions: exactly one z_{ik} equals 1 and the others are 0, so raising each component's candidate factor to the power z_{ik} keeps the right one and turns every other into $[\,\cdot\,]^0 = 1$:

$$p(\mathbf{x}_i, z_i;\, \theta) = \prod_{k=1}^{K} \big[\pi_k\, \varphi(\mathbf{x}_i;\, \boldsymbol{\mu}_k, \Sigma_k)\big]^{z_{ik}}.$$

The pairs being i.i.d., the complete-data likelihood is the product of these joint densities over the sample,

$$L_c(\theta) = \prod_{i=1}^{n} p(\mathbf{x}_i, z_i;\, \theta) = \prod_{i=1}^{n} \prod_{k=1}^{K} \big[\pi_k\, \varphi(\mathbf{x}_i;\, \boldsymbol{\mu}_k, \Sigma_k)\big]^{z_{ik}}, \tag{21}$$

and taking the logarithm turns both products into sums and each exponent into a factor:

$$\ell_c(\theta) = \log L_c(\theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} z_{ik} \big[\log \pi_k + \log \varphi(\mathbf{x}_i;\, \boldsymbol{\mu}_k, \Sigma_k)\big]. \tag{22}$$

Compare (22) with the observed-data log-likelihood (20): no log-of-sum anywhere. The sum over k sits *outside* the logarithm, because the indicators have already decided who belongs where. Maximizing ℓ_c is chapter 1's problem K times over: with the counts $n_k = \sum_i z_{ik}$, the maximizer is $\hat\pi_k = n_k/n$, $\hat{\boldsymbol{\mu}}_k$ the mean of group k's points, and $\widehat{\Sigma}_k$ their scatter. This is the split-and-fit estimation of section 2.1, in formulas.

The E-step: expected indicators. The indicators are missing. EM's move is the simplest one available: replace each missing z_{ik} by its *conditional expectation* given what we do observe, under the current parameter guess $\theta^{(t)}$. Since z_{ik} is a 0/1 variable, its expectation is a probability, and one we have already computed:

$$\mathbb{E}\big[z_{ik} \mid \mathbf{x}_i;\, \theta^{(t)}\big] = \mathbb{P}\big(Z_i = k \mid \mathbf{x}_i;\, \theta^{(t)}\big) = \gamma_{ik}^{(t)},$$

the responsibilities of section 2.2. Because ℓ_c is *linear* in the z_{ik}, substituting expectations for indicators gives exactly the expected complete-data objective

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}\big[\ell_c(\theta) \mid \mathbf{x}_1, \ldots, \mathbf{x}_n;\, \theta^{(t)}\big] = \sum_{i=1}^{n} \sum_{k=1}^{K} \gamma_{ik}^{(t)} \big[\log \pi_k + \log \varphi(\mathbf{x}_i;\, \boldsymbol{\mu}_k, \Sigma_k)\big]. \tag{23}$$

The M-step: fractional counts. Now maximize Q over θ. Compare Q with ℓ_c: they are the *same function*, with the hard indicators $z_{ik} \in \{0, 1\}$ replaced by the soft weights $\gamma_{ik} \in [0, 1]$. The maximization therefore goes through verbatim, with *fractional counts* in place of integer ones: writing $n_k = \sum_i \gamma_{ik}$ for component k's effective number of points,

$$\pi_k^{new} = \frac{n_k}{n}, \qquad \boldsymbol{\mu}_k^{new} = \frac{1}{n_k} \sum_{i=1}^{n} \gamma_{ik}\, \mathbf{x}_i, \qquad \Sigma_k^{new} = \frac{1}{n_k} \sum_{i=1}^{n} \gamma_{ik}\, (\mathbf{x}_i - \boldsymbol{\mu}_k^{new})(\mathbf{x}_i - \boldsymbol{\mu}_k^{new})^\top \tag{24}$$

that is, group proportions together with chapter 1's means and scatters, each observation counted in every group according to its responsibility. Given the decision to replace the missing indicators by their expectations, the updates are forced rather than postulated. The algorithm is now just the alternation of these two computations:

Definition 2.3.1 (EM for Gaussian mixtures)

Initialize $\theta^{(0)}$; alternate until convergence:

- *E-step*: given $\theta^{(t)}$, compute the responsibilities $\gamma_{ik} = \mathbb{P}(Z_i = k \mid \mathbf{x}_i; \theta^{(t)})$ and form $Q(\theta \mid \theta^{(t)})$;
- *M-step*: set $\theta^{(t+1)} = \arg\max_\theta Q(\theta \mid \theta^{(t)})$ (the fractional-count updates above).

What makes EM more than a plausible heuristic is that each round provably improves the actual objective:

Theorem 2.3.1 (Monotonicity of EM)

The observed-data log-likelihood never decreases along EM iterations: $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$.

Proof. Fix $\theta^{(t)}$ and write $q_{ik} = \gamma_{ik}^{(t)}$ for the E-step responsibilities, which satisfy $q_{ik} > 0$ and $\sum_k q_{ik} = 1$. For any θ, insert q_{ik}/q_{ik} inside each inner sum of ℓ and apply Jensen's inequality to the concave logarithm ($\log \mathbb{E} \geq \mathbb{E} \log$ for the distribution $q_{i\cdot}$):

$$\ell(\theta) = \sum_{i=1}^{n} \log \sum_{k=1}^{K} q_{ik} \frac{\pi_k \varphi(\mathbf{x}_i; \boldsymbol{\mu}_k, \Sigma_k)}{q_{ik}} \;\geq\; \sum_{i=1}^{n} \sum_{k=1}^{K} q_{ik} \log \frac{\pi_k \varphi(\mathbf{x}_i; \boldsymbol{\mu}_k, \Sigma_k)}{q_{ik}} \;=\; Q(\theta \mid \theta^{(t)}) + H,$$

where Q is the expected complete-data objective defined above and $H = -\sum_{i,k} q_{ik} \log q_{ik}$ does not depend on θ. Two facts finish the proof. First, at $\theta = \theta^{(t)}$ the bound is an *equality*: plugging the definition of γ_{ik}, each ratio $\pi_k \varphi(\mathbf{x}_i)/q_{ik}$ equals the mixture density $f(\mathbf{x}_i; \theta^{(t)})$, a constant in k, and Jensen is tight on constants. Second, the M-step maximizes Q by construction (the exact updates (24)). Chaining the two:

$$\ell(\theta^{(t+1)}) \;\geq\; Q(\theta^{(t+1)} \mid \theta^{(t)}) + H \;\geq\; Q(\theta^{(t)} \mid \theta^{(t)}) + H \;=\; \ell(\theta^{(t)}). \qquad \square$$

The proof uses the M-step only through $Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)})$. The regularized covariance updates of section 2.4 are not the exact maximizer of Q, so for them the guarantee holds only as long as each update still increases Q. In the run of figure 2.3, which uses such an update, the likelihood does increase at every iteration.

Figure 2.3 shows the algorithm at work on the customer data: from a poor random start ($\ell = -4683$), a single round of soft reassignment already reshapes the components ($\ell = -4508$), and by iteration 25 the three ellipses have locked onto the groups ($\ell = -3720$), with the points sitting between groups shared between components, in blended colors, rather than assigned to one group.

2.4 What can go wrong?

Monotone improvement is a strong property, but it is a *local* guarantee, and fitted mixtures fail in reproducible, avoidable ways.

Local maxima and initialization. Theorem 2.3.1 promises only that the likelihood never decreases. At best EM settles at a stationary point of ℓ, typically a local maximum, and mixture likelihoods have many (different clusterings can each be locally optimal); at worst it runs off toward the degenerate spikes described next. Where EM lands depends entirely on $\theta^{(0)}$. Standard practice runs EM from several initializations (random data points as initial means, or the K-means solution of section 2.7) and keeps, among the runs that did not degenerate, the one with the best final likelihood, stopping each run when ℓ stalls between iterations.

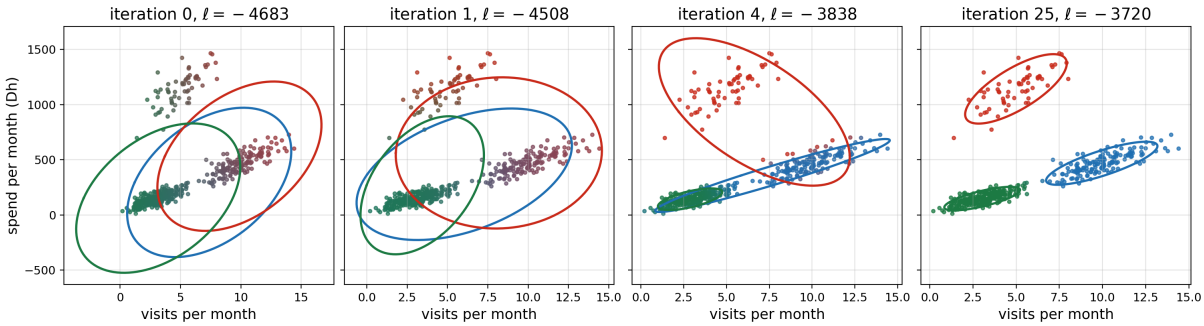


Figure 2.3: One EM run ($K = 3$) on the customers cloud: point colors blend the component colors in proportion to responsibilities; ellipses are the current component covariances. The log-likelihood increases at every step, as theorem 2.3.1 promises for the exact M-step (the run adds the tiny ridge of section 2.4).

The likelihood is unbounded. A second failure is more serious: the global maximum of the likelihood *does not exist*. Center one component on a single data point, $\boldsymbol{\mu}_1 = \mathbf{x}_1$, and let $\Sigma_1 \to 0$: that component's density at $\mathbf{x}_1$ explodes while every other point remains covered by the remaining components, so $\ell(\theta) \to +\infty$. This behavior, called *component collapse*, occurs in practice: a run whose component captures one or two points can enter this regime, its likelihood increasing while the component degenerates into a spike around a single observation. What we seek is a good *local* maximizer away from these degeneracies. Figure 2.4 shows the pull toward collapse on a crescent-shaped cloud whose curved geometry no small-K mixture fits well, inviting many components. As K grows, the fitted density wraps the sample ever more tightly and the likelihood climbs at every step. At $K = 12$ the improvement is no longer geometric: several components have contracted onto a handful of nearly aligned points and flattened onto the line through them, visible as needle-thin ridges of the density. These are the first steps of the unbounded path above: a component's likelihood gain grows like $-d \log \sigma$ when it shrinks to a point spike of scale σ, and like $-\log \sigma_{\min}$ when only its smallest standard deviation $\sigma_{\min}$ shrinks, as for these needles.

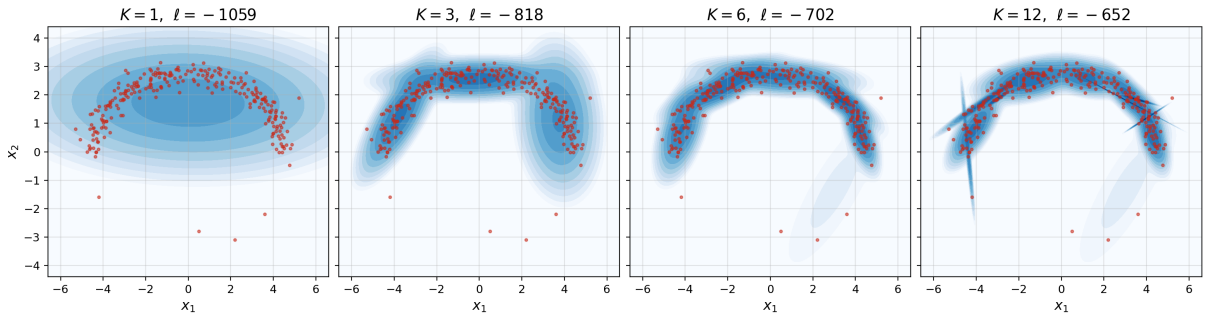


Figure 2.4: Fitted mixture density (blue, log scale) on a crescent-shaped cloud for a growing number of components. The log-likelihood increases at every step. At $K = 12$ several components have contracted onto a few points (the needle-thin ridges), and further gains come from sharpening these spikes, not from fitting the shape of the cloud: the onset of component collapse.

Singular covariances and regularization. Short of full collapse, a component whose effective sample $n_k = \sum_i \gamma_{ik}$ is not much larger than d produces an M-step covariance as ill-conditioned as any sample covariance estimated from too few points: this is chapter 1's covariance-estimation problem again, now per component; each Σ_k is estimated from effectively n_k observations, and there are K of them. The remedy is also chapter 1's, applied inside the M-step: *regularize the estimator* by blending each updated covariance $\widehat{\Sigma}_k$ (the update (24)) with a fixed multiple of the

identity,

$$\Sigma_k \longleftarrow (1-\lambda)\,\widehat{\Sigma}_k + \lambda\,\tau\,\mathrm{I}_d, \qquad \tau = \mathrm{tr}(\widehat{\Sigma}_{\mathrm{MLE}})/d, \tag{25}$$

for a small $\lambda \in (0, 1]$, where $\widehat{\Sigma}_{\mathrm{MLE}}$ is chapter 1's sample covariance of the whole dataset; the scale τ, the average variance of the data, is computed once before EM starts and never changes. The eigenvectors of $\widehat{\Sigma}_k$ are untouched and every eigenvalue is pulled toward τ, with a floor at $\lambda\tau > 0$ that no iteration can lower: the matrix stays invertible, each component density stays below $(2\pi\lambda\tau)^{-d/2}$, and the collapse $\sigma \to 0$ becomes *impossible by construction*. The target scale has to be fixed: a scale computed from $\widehat{\Sigma}_k$ itself would shrink with the collapsing component and remove the floor exactly when it is needed. Standard EM implementations apply a version of this; ours adds a fixed ridge $10^{-6}\,\mathrm{I}_d$ to each M-step covariance, a tiny instance of the same fixed floor (tiny enough that the needles of figure 2.4 formed with it in place).

2.5 Covariance structures

The parameter bill. The full mixture is expensive. Counting free parameters: $K-1$ weights, Kd mean coordinates, and one symmetric covariance per component (the dominant term), $d(d+1)/2$ entries each:

$$p_K = (K-1) + Kd + K\,\frac{d(d+1)}{2}. \tag{26}$$

For $d = 50$ and $K = 10$ that is $9 + 500 + 12\,750 = 13\,259$ parameters, to be estimated from unlabeled data in which each component effectively sees only its share n_k of the observations. This is the regime where the failures of section 2.4 occur.

Constrained variants. The structural response is chapter 1's other move, *regularize the problem*: restrict the covariance structure, trading cluster expressiveness for parameter economy. From the most to the least expensive in general (shared is cheaper than diagonal when $K > (d+1)/2$, dearer otherwise):

- *full*: one free Σ_k per component ($K\,d(d+1)/2$ covariance parameters); arbitrary shape and orientation per cluster;
- *shared* (tied): $\Sigma_k = \Sigma$ for all k ($d(d+1)/2$); all clusters have the same shape, only their centers differ;
- *diagonal*: $\Sigma_k = \mathrm{diag}(\sigma_{k1}^2, \ldots, \sigma_{kd}^2)$ (Kd); axis-aligned ellipsoidal clusters, no within-cluster correlations;
- *spherical*: $\Sigma_k = \sigma_k^2 \mathrm{I}_d$ (K); round clusters of per-component radius.

Every variant keeps the same EM algorithm: only the M-step's covariance update changes, maximizing Q under the constraint (for the shared variant, a responsibility-weighted pooling across components; for the diagonal one, coordinatewise variances). The choice is a modeling decision of the kind flagged in chapter 1: the constrained models gain stability on small groups at the cost of expressiveness (a spherical model cannot represent an elongated, tilted cluster). How to make the choice, and how to choose K itself, is the next section's question.

2.6 Choosing the number of components

The number of components cannot be chosen by the likelihood. Since the likelihood is unbounded, write $\hat{\theta}_K$ for the best non-degenerate EM fit with K components. The models are nested (a mixture with $K+1$ components can imitate any K-component one by splitting one component into two identical halves of weight $\pi_k/2$ each), so the best attainable fit $\ell(\hat{\theta}_K)$ does not decrease

with K, and selecting the best-fitting K would always answer "as many as possible". This is the first appearance of a distinction that recurs throughout the course: a better fit is not a better model. A mixture with one component per data point fits the sample arbitrarily well while describing nothing beyond it; this failure is called *overfitting*: a flexible model absorbing the sampling noise of the particular dataset. The opposite failure, a model too rigid to represent real structure, is *underfitting*; a single Gaussian forced onto the three-group customer cloud is an example. The standard remedy penalizes the parameter count p_K of section 2.5: the *Bayesian information criterion* [3]

$$\mathrm{BIC}(K) = -2\,\ell(\hat{\theta}_K) + p_K \log n \tag{27}$$

is minimized over K. Figure 2.5 plots both terms on the customer data ($d = 2$, $n = 470$, three latent groups). The fit term $-2\,\ell(\hat{\theta}_K)$ falls steeply while real structure is being captured (9201 at $K = 1$, 7831 at $K = 2$, 7440 at $K = 3$), then decreases only slightly (7429, 7415, 7410): each extra component improves the fit marginally, by modeling sampling noise, so the fit curve alone cannot select K. BIC sits above it by the penalty $p_K \log n$, the growing gap between the two curves. The penalty grows linearly in p_K while the fit gains shrink, so BIC attains its minimum 7544 at $K = 3$ and increases thereafter, the number of latent groups in the data.

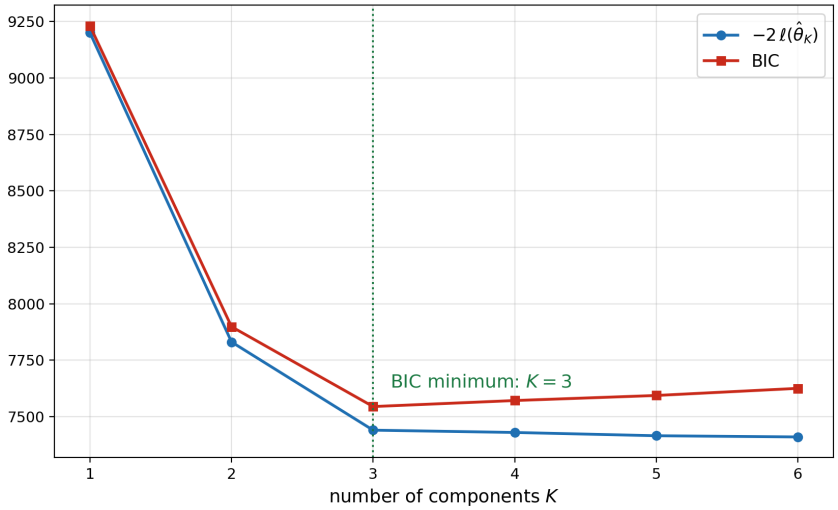


Figure 2.5: Model selection on the customers cloud (three latent groups). The fit term $-2\,\ell(\hat{\theta}_K)$ decreases forever; BIC adds the penalty $p_K \log n$ (the widening gap) and turns into a minimum at $K = 3$.

The same criterion applies to the covariance variants, whose p_K differ precisely as section 2.5 counted: BIC compares them as it compares values of K. It does not apply to the regularization λ of section 2.4, which leaves p_K unchanged: BIC then reduces to the likelihood, which favors $\lambda \to 0$. The weight λ is instead chosen by the likelihood of held-out data (cross-validation), a score that also serves for K and the covariance structure when data are plentiful. BIC balances fit against complexity without setting data aside: when the sample is too scarce to hold part of it out, the penalty replaces the held-out score.

2.7 GMM vs K-means

The best-known clustering algorithm, *K-means*, predates EM and underlies much of this machinery: pick K centers, assign each point to its nearest center, recompute each center as the mean of its points, repeat. It is exactly what EM degenerates into under three drastic simplifications: force all covariances to $\sigma^2 \mathrm{I}_d$, all priors to $1/K$, and let $\sigma \to 0$. The responsibilities then collapse to hard 0/1 assignments to the nearest center, and the M-step becomes the plain mean.

K-means, in other words, is a Gaussian mixture that has silently assumed *spherical, equally-sized, equally-weighted* clusters.

The comparison is again a matter of assumptions. K-means is fast, simple and uses few parameters; when its hidden assumptions hold, it is hard to beat. When they fail, the algorithm gives no warning: figure 2.6 shows two elongated, tilted clusters. K-means, blind to shape, cuts *across* both clusters and misassigns 27% of the points, while the full-covariance GMM recovers the two groups almost perfectly (0.8%). The price of that fidelity was computed in section 2.5: each full covariance costs $d(d+1)/2$ parameters, so in high dimension the GMM needs data that K-means does not; the constrained variants of section 2.5 populate the spectrum between the two extremes.

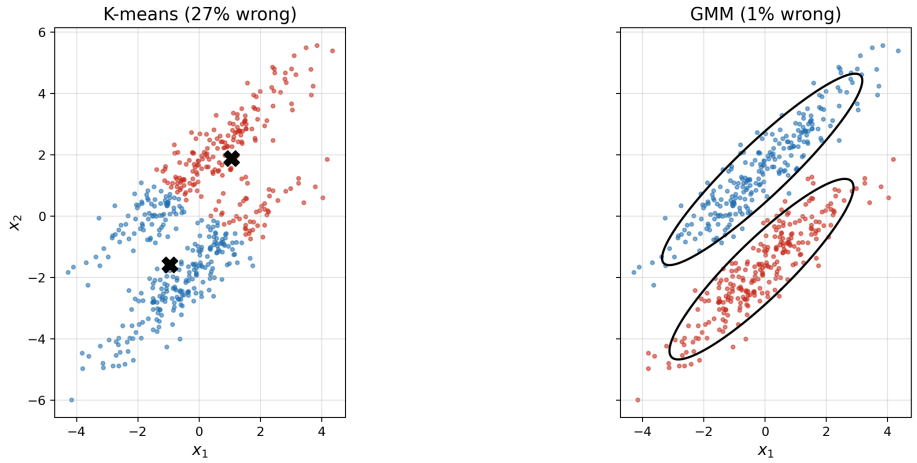


Figure 2.6: Two elongated correlated clusters. K-means (left) assumes spherical clusters and slices across the true groups; the full-covariance GMM (right) models the shape and recovers them.

2.8 Interpreting the fitted components

A mixture has now been fitted by EM, with restarts, a sensible covariance structure and a BIC-chosen K. Two cautions are needed before its components are interpreted.

Label switching. The caveat recorded in section 2.2 can now be stated in full. Permuting the components (relabeling $(\pi_1, \boldsymbol{\mu}_1, \Sigma_1)$ as component 2 and vice versa) leaves the mixture density f strictly unchanged, so the parameters are identifiable only *up to relabeling*: a K-component mixture has $K!$ equivalent parameterizations. For the clustering itself this is harmless (the *partition* of the data does not depend on the numbering), but it matters the moment components are tracked across runs: "component 1" of one EM run may be "component 2" of the next, and averaging parameters across bootstrap replicates, or comparing "component 1" between two subsamples, is meaningless until the components have been matched (by their means, or by the overlap of their responsibilities).

Components are not always clusters. The model fits a density; the scientific question is usually about subpopulations, and the two need not coincide. A single skewed or heavy-tailed cluster is often fitted by *two* Gaussian components sharing the work: BIC may rightly prefer $K = 3$ on data with two "real" groups, one of them non-Gaussian. The safe reading is that components are building blocks of a density estimate, responsibilities are calibrated membership probabilities under that estimate, and any claim that "component k is a real subtype" (three customer types, two cell subtypes) needs evidence beyond the fit: stability across restarts and subsamples, and ideally external validation.

What the fitted mixture delivers. With those cautions in place, return to the prologue's five purposes of a fitted distribution, on the customer data that opened the chapter. *Typicality*: the mixture reports three typical customers (one per segment, each with its own spread) instead of one average customer who resembles nobody in the file. *Dependence*: within-group covariances now describe how visits and spend co-vary *inside* a segment, no longer confounded with the differences between segments. *Anomaly*: thresholding the *mixture* density flags customers implausible under *every* segment, and chapter 1's chi-squared calibration applies within each component. *Prediction*: a missing spend entry is imputed by a responsibility-weighted blend of the per-segment distributions of spend given visits, each one chapter 1's conditional distribution of a Gaussian vector. *Generation*: synthetic customers are drawn group-first (pick a segment, then a customer from it) and reproduce three clouds rather than one average blur. One family change, motivated by one picture, repaired all five answers.

References

[1] K. Pearson. Contributions to the mathematical theory of evolution. *Philosophical Transactions of the Royal Society of London A*, 185:71–110, 1894.

[2] A. P. Dempster, N. M. Laird and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society, Series B*, 39(1):1–38, 1977.

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Chapter 3
Hypothesis testing revisited

Every chapter so far has ended the same way: a model was written down and its parameters were estimated, and in chapter 1 the estimates came with a measure of their own uncertainty. This chapter turns that machinery around. Instead of asking *what value the data suggest for a parameter*, it asks whether the data support or undermine a *claim*: that a player is better than another, that a coefficient is zero, that a die is fair. You have met the answer's vocabulary in first year: null hypothesis, p-value, level, rejection. The purpose here is to rebuild that vocabulary from its starting point, a single probability computation performed three centuries ago, and to assemble, step by step, the reasoning ladder that every statistical test follows, including the ones in the chapters ahead. The running example is two friends and a football video game.

3.1 Eighty-two years of boys

The London Bills of Mortality recorded yearly christenings, boys and girls separately, through the seventeenth and early eighteenth centuries. Over the 82 consecutive years from 1629 to 1710, every single year recorded more boys christened than girls [1]. What is the probability of that?

The reflex is to compute. Treat each year as a coin flip (boys or girls in the majority, each with probability 1/2) and treat the years as independent. Then

$$\mathbb{P}(82 \text{ boy years out of } 82) = \left(\tfrac{1}{2}\right)^{82} \approx 2.1 \times 10^{-25}. \tag{28}$$

The computation is correct, and it does not answer the question that was asked. It answers a different one: *if* each year were a fair, independent coin flip, what would the chance of the record be? The two assumptions were loaded silently. Where did the 1/2 come from? Nothing in the records says that boy-majority and girl-majority years are equally likely; we supplied that. Who said the years are independent? We did, without noticing. The number 2.1×10^{-25} is not a fact about London; it is a fact about a *model* of London that we chose in the act of computing.

That observation leaves exactly two readings. Either an event of probability 10^{-25} happened (an event so rare that no human institution has ever recorded one), or at least one of the loaded assumptions is false. John Arbuthnot, the physician who published this computation in 1710, took the second reading: the record is overwhelming evidence *against* the fair-coin model of births. This inversion is the founding move of statistical testing. A probability computed *under* an assumption is turned into evidence *about* the assumption: assume, compute the probability of what was observed, and let a tiny result testify against the assumption package.

The computation has a limit that is worth stating before the machinery makes it precise. It cannot say *which* part of the package fails. Perhaps boys are genuinely more probable at birth (they are, slightly); perhaps the years are not independent; the number 2.1×10^{-25} indicts the package as a whole and is silent about the parts. Arbuthnot himself overreached on this point: he read the rejection as proof of a specific cause (in his case, divine design), when the calculation licenses only "not chance, as chance was modeled here." Keeping honest books (what is assumed, what is claimed, and what the computed number does and does not say) is exactly what the formal machinery of this chapter is for.

3.2 Who's the better gamer ?

This section turns Arbuthnot's inversion into a method, on a dispute closer to home, in five steps that the rest of the chapter extends: pose the question, translate it into hypotheses, compress the evidence into a statistic, derive that statistic's distribution under the null, and turn a tail probability into a decision with known error rates.

3.2.1 A question about a game

Karim and Yassine, two students of this course, have played FIFA against each other all semester. Karim has won 14 of the last 20 games and claims to be, simply, the better player. Yassine's answer is one word: luck. The dispute is real, and it is decidable, but not by staring at the score. Fourteen wins out of twenty is perfectly possible between equal players, and it is also what a genuinely better player might produce. The question that data can answer is Arbuthnot's question, asked deliberately this time: *if the two were equally good, how surprising would* 14 *out of* 20 *be?*

To ask it cleanly, give the assumption a name instead of smuggling it in. Let θ denote the probability that Karim wins a single game, and model the 20 games as independent, each won by Karim with probability θ.[1] The dispute becomes a statement about θ:

$$H_0 : \theta = \tfrac{1}{2} \qquad \text{against} \qquad H_1 : \theta > \tfrac{1}{2}. \tag{29}$$

Definition 3.2.1 (Null and alternative hypotheses)

A statistical hypothesis is a statement about the parameter of a model. A test opposes a *null hypothesis* H_0, the statement to be challenged, to an *alternative hypothesis* H_1, the claim that would replace it. In (29) the null is *simple*: it specifies the parameter completely. A null covering a whole set of values is *composite* (section 3.2.5). The alternative here is *one-sided*: it points in one direction.

Two remarks on the casting of roles. First, why exact equality, when Yassine's real claim is broader: Karim is *not better*, that is $\theta \leq 1/2$? Because computation needs a pinned-down model, and neither the full claim nor the alternative provides one: "not better" allows every θ up to $1/2$, "better" every θ above it. The boundary value $\theta = 1/2$ is the one point that does pin the model down (as the implicit $1/2$ per year did in section 3.1), and it is the right point to single out: of all the values "not better" allows, $1/2$ produces lopsided scores most easily, so a rejection calibrated there rejects the broader claim as a whole (sections 3.2.5 and 3.3.3 make this precise). Second, the two hypotheses will not be treated symmetrically. Everything that follows measures evidence *against* H_0; the precise sense in which the null can be rejected but never accepted is the destination of section 3.2.5.

3.2.2 A statistic and its null distribution

The data are twenty game results. To weigh them, they must first be compressed into a single number that measures the discrepancy from H_0 in the direction of the claim. Such a number is called a *test statistic*. Here the choice is obvious: the number of wins. (Arbuthnot's statistic was the number of boy-majority years, 82 out of 82, chosen without ever being named.) A good statistic has two properties: it is computable from the data alone (no unknown parameter in its formula), and its distribution under H_0 is known, or at least obtainable.

For the number of wins the null distribution comes from one line. Write $S = \sum_{i=1}^{20} W_i$, where W_i indicates a Karim win in game i. Under H_0 the W_i are independent Bernoulli variables with

[1] The letter θ, not p, on purpose: throughout this chapter p is reserved for the p-value, defined in section 3.2.3.

parameter 1/2, so

$$\text{under } H_0: \quad S \sim \mathrm{Bin}\left(20, \tfrac{1}{2}\right), \qquad \mathbb{P}(S = k) = \binom{20}{k}\left(\tfrac{1}{2}\right)^{20}. \tag{30}$$

This distribution, the law of the statistic in the world where the null is true, is called the *null distribution*, and it is the measuring device of the whole chapter: it says which values of S are ordinary under H_0 and which are remarkable.

There is a second route to the same object. Instead of the formula, *simulate* the world of H_0: make a computer play 100 000 fake seasons of 20 fair coin flips each and histogram the number of wins. Figure 3.1 shows the result on top of the exact distribution (30): the two agree, as they must. The formula is faster and exact; the simulation needs nothing but the ability to generate data under the null, a route that stays open even for statistics whose null distribution admits no usable formula.

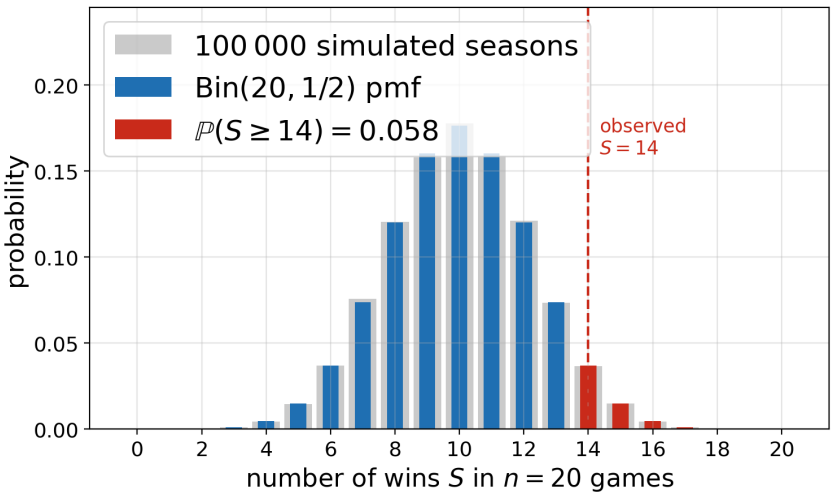


Figure 3.1: The null distribution of the number of wins S in 20 games: the exact $\mathrm{Bin}(20, 1/2)$ probabilities (blue) and the histogram of 100 000 simulated fair seasons (gray) agree. The observed $S = 14$ sits in the right tail; the red bars are the outcomes at least as favorable to Karim, of total probability 0.058.

Nothing in this section is new probability. A model, a statistic, a sampling distribution under the model: this is chapter 1's machinery in miniature, with one Bernoulli parameter in place of a mean vector and a covariance. The new idea is the direction of use. Estimation read the sampling distribution forward, to attach uncertainty to $\hat{\theta}$; testing reads it backward, as a scale on which the observed value of the statistic is judged for surprise.

3.2.3 The p-value

The observed statistic is $S = 14$. On the null distribution (30), how remarkable is it? The relevant region is not the single value 14 but everything at least as favorable to the claim, because the question is directional: the alternative in (29) points toward large S. Summing the tail of (30),

$$\mathbb{P}(S \geq 14) = 0.058:$$

about one twenty-game season in seventeen between *equal* players produces a score at least this lopsided. This tail probability is the test's summary of the evidence and has a name.

Definition 3.2.2 (p-value)

The *p-value* of an observed statistic s_{obs} is the probability, computed under the null distribution, of a statistic at least as extreme as the one observed, where "extreme" means *in the direction of the alternative*:

$$p \;=\; \mathbb{P}_{H_0}(S \geq s_{\text{obs}}) \qquad \text{(here, for } H_1 : \theta > \tfrac{1}{2}\text{)}. \tag{31}$$

A small p-value means: a world where H_0 holds rarely produces data this favorable to H_1.

Set the two computations of this chapter side by side. Arbuthnot's p-value, 2.1×10^{-25}, leaves no doubt. Karim's, 0.058, is not small enough to rule out luck and not large enough to dismiss the claim. Deciding in such cases requires a rule fixed in advance.

3.2.4 The level and the threshold

The rule is fixed *before* looking at the data. Choose a *level* α (the convention in most fields is 5%) and decide as follows: reject H_0 when the p-value is at most α. Equivalently, since $p \leq \alpha$ exactly when the statistic falls beyond the corresponding quantile of the null distribution, the rule can be written on the scale of S itself:

$$\text{reject } H_0 \iff p \leq \alpha \iff S > q_{1-\alpha}, \tag{32}$$

where $q_{1-\alpha}$ is the $(1-\alpha)$-quantile of the null distribution (the smallest q with $\mathbb{P}_{H_0}(S \leq q) \geq 1-\alpha$), read off in the direction the alternative points to; the statistic must strictly exceed it. The threshold is not a new object: it is a quantile of the null law, and the alternative decides *which* quantile (upper here, lower or two-sided for other alternatives; see section 3.3). For $\text{Bin}(20, 1/2)$ at $\alpha = 5\%$, the quantile is $q_{0.95} = 14$, and the tail probabilities are $\mathbb{P}(S \geq 14) = 0.058$ and $\mathbb{P}(S \geq 15) = 0.021$: the rejection region is $S > 14$, that is $S \geq 15$. Karim's 14 falls one game short, and the test does not reject.

One feature of this example repays attention. No rejection region of the form $S \geq k$ has probability exactly 5% under $\text{Bin}(20, 1/2)$: the choice is between 5.8% (too lax) and 2.1% (the real size of the test actually used). A discrete statistic offers only a staircase of achievable levels, and the convention is to step down, so the guaranteed level is conservative. Exact-5% rejection regions are a luxury of continuous statistics.

3.2.5 Type I and Type II Errors and the Power of a Test

Return to the rule of section 3.2.4: reject when $S \geq 15$. Its properties can be computed before they are named. If the players are equal, $\theta = 0.5$, the rule rejects with probability $\mathbb{P}_{0.5}(S \geq 15) = 0.021$. If Karim actually wins 60% of his games, it rejects with probability $\mathbb{P}_{0.6}(S \geq 15) = 0.13$; if he wins 70%, with probability $\mathbb{P}_{0.7}(S \geq 15) = 0.42$. Nothing restricts the computation to these three values. The rule's entire behavior is captured, one parameter value at a time, by the function

$$r(\theta) \;=\; \mathbb{P}_{\theta}(S \geq 15), \qquad \theta \in [0, 1], \tag{33}$$

its *rejection probability*. Figure 3.2 draws r over the whole interval; the parameter is unknown but not random, and each point of the curve is a probability computed in the model with θ fixed at that value.

Reading the figure brings back the full form of the dispute, the one section 3.2.1 set aside for the sake of computation: Karim is *not better*. As sets of parameter values, the hypotheses are

$$H_0 : \theta \leq \tfrac{1}{2} \quad (\Theta_0 = [0, \tfrac{1}{2}]), \qquad H_1 : \theta > \tfrac{1}{2} \quad (\Theta_1 = (\tfrac{1}{2}, 1]),$$

the two regions of the figure. The single value $\theta = 1/2$ that every computation so far ran on is the boundary of Θ_0, and it is no loss: r is increasing in θ (more skill, more wins), so among all the

values the null allows, the boundary is where the rejection probability is largest. (Section 3.3.3 makes this precise.)

The figure now reads as follows. Over Θ_0 the height of the curve is the probability of a false alarm: rejecting although Karim is not better. Over Θ_1 the same height is the probability of a justified alarm: rejecting because he is. A good rule keeps the first small and the second large: a statistical test controls the probability of rejection wherever the null is true, while trying to make that same probability as large as possible wherever the alternative is true. Every term defined in the rest of this subsection is a reading of r on one of the two regions.

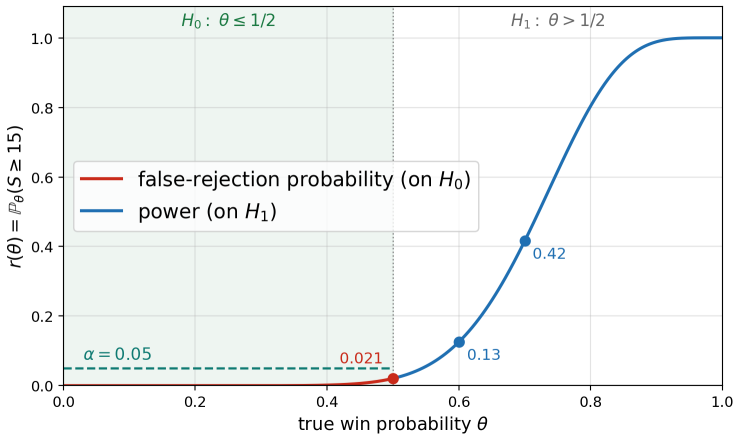


Figure 3.2: The rejection probability $r(\theta) = \mathbb{P}_\theta(S \ge 15)$ of the rule "reject when $S \ge 15$" ($n = 20$), over the whole parameter range. Over the shaded null region its height is a false-rejection probability, largest at the boundary: $r(1/2) = 0.021$, below the α line. Over the alternative region the same height is the power: 0.13 at $\theta = 0.6$, 0.42 at 0.7.

Definition 3.2.3 (Type I error, size, level)

A *type I error* is the event of rejecting H_0 while the true parameter lies in Θ_0; at a given null value θ, it has probability $r(\theta) = \mathbb{P}_\theta(\text{reject } H_0)$. The *size* of a test is its largest false-rejection probability over the null,

$$\text{size} = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\text{reject } H_0), \tag{34}$$

and the test *has level* α when its size is at most α.

For the wins rule, the monotonicity of r places the size at the boundary: size $= r(1/2) = 0.021 \le 0.05$. The rule has level 5% over the whole null region, and not by luck: the threshold of section 3.2.4 was a quantile of the boundary null distribution, chosen precisely so that the size stays under α whatever the data turn out to be.

The power. Over the alternative region the heights of r measure what the test is worth, and get their own name:

$$\pi(\theta) = \mathbb{P}_\theta(\text{reject } H_0), \qquad \theta \in \Theta_1, \tag{35}$$

the *power function*. Power is a function, not one number: detecting a 51%, a 60% and a 70% player are very different tasks. The wins rule handles them poorly. If Karim truly wins 60% of games, the test detects it with probability 0.13; at a true 70%, probability 0.42; only around 80% does detection become likely at 0.80. Twenty games are simply not much information; detecting a 60% player with power 0.80 takes about 160 games (approximately: for a discrete test the power is not monotone in n).

Figure 3.3 draws the kind of curve such a rule aims for: still under the α line on all of Θ_0, then rising fast once θ crosses into Θ_1. The rise can be steep but never a jump: at fixed n the

rejection probability of the wins rule is a polynomial in θ, hence continuous, and what more data buy is exactly a steeper S-curve (the 160-game figure above is one point of that sharpening). This is the design principle Neyman and Pearson formalized in 1933 [2]: fix the level first, then, among all tests satisfying that level constraint, seek the largest possible power under the alternatives.

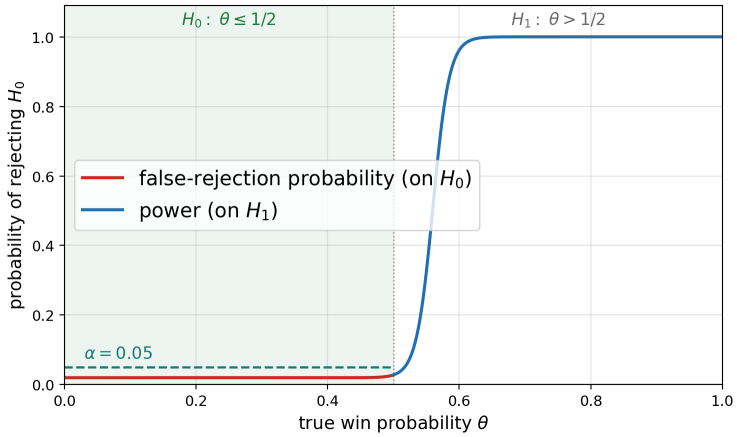


Figure 3.3: A rejection curve one would wish for, on the same axes: below the α line throughout the null region, close to one throughout most of the alternative. At fixed n the rejection probability is continuous in θ, so the steep rise can never be a jump; larger n steepens it.

The type II error. One failure remains: the test stays silent although Karim really is better. At an alternative value θ, its probability is the complement of the power,

$$\beta(\theta) \;=\; \mathbb{P}_\theta(\text{do not reject } H_0) \;=\; 1 - \pi(\theta), \qquad \theta \in \Theta_1. \tag{36}$$

No dial of the construction controls it; it is whatever the level and the sample size leave over. Read off figure 3.2: $\beta(0.6) = 0.87$ and $\beta(0.7) = 0.58$: at twenty games, a real edge is missed more often than it is caught.

The summary. Crossing what is true with what the test decides:

	H_0 true (Karim not better)	H_1 true (Karim better)
reject H_0	**type I error** (prob. $r(\theta)$; size $\le \alpha$)	correct (prob. $\pi(\theta)$)
do not reject	correct	**type II error** (prob. $\beta(\theta)$)

These numbers give the failure to reject its correct meaning. Karim's season, $S = 14$ and $p = 0.058$, is compatible with equal skill, *and* compatible with Karim winning 70% of all games, an edge the test misses more often than it catches. Not rejecting H_0 therefore establishes nothing about H_0: the data were too few, or the effect too small, for the claim to clear the bar. The test measures evidence against the null and never for it; **failure to reject H_0 means insufficient evidence against it, not evidence that it is true**. This is the asymmetry announced in section 3.2.1, and it is the single most common point of confusion in applied work: *absence of evidence is not evidence of absence.*

Key idea

The testing ladder. Every hypothesis test in this course follows the same basic logic.

1. **Start from the question.** What scientific or practical claim is being assessed? (Is Karim better?)

2. **State H_0 and H_1.** Translate the question into a null hypothesis to test and an alternative representing the claim of interest.
3. **Choose a statistic.** Reduce the data to a quantity that becomes more extreme when the data depart from H_0 in directions relevant to H_1.
4. **Determine its behavior under H_0.** Derive or simulate the distribution of the statistic when the null hypothesis holds.
5. **Interpret the p-value.** If H_0 were true, how probable would a statistic at least as extreme as the one observed be?
6. **Report effect size and uncertainty.** Statistical significance alone does not say how large or important the effect is (section 3.4).
7. **Check assumptions and design issues.** How sensitive is the conclusion to sample size, distributional shape, dependence, or multiple testing? (Section 3.5 works out the many-tests case.)
8. **Conclude in context.** Return to the original question: what evidence do the data provide, under the assumptions of the test?

Steps 1–5 have now been walked on the FIFA data; steps 6–8 are the rest of this chapter.

3.3 The alternative H_1 decides the test

Everything computed so far (the tail, the quantile, the p-value) was read off in the direction of $H_1 : \theta > 1/2$. This section varies the alternative and the null, and once the data, to make one point three times: the alternative, chosen from the question and not from the data, decides what "extreme" means, hence the rejection region, hence the p-value.

3.3.1 One-sided or two-sided

Yassine, unwilling to concede the framing, restates the dispute: "one of us is better, who knows which." The claim is now $H_1 : \theta \neq 1/2$, and both directions count as extreme. The p-value collects both tails: on the same 14 wins,

$$p = \mathbb{P}(|S - 10| \geq 4) = 0.115,$$

double the one-sided 0.058, by the symmetry of $\mathrm{Bin}(20, 1/2)$ (figure 3.4). The same twenty games, summarized by the same statistic, yield 0.058 or 0.115 depending only on the question asked.

Remark

The alternative is chosen before the data. The direction of H_1 must come from the question (is the claim "Karim is better" or "the players differ"?) and must be fixed before the results are seen. Choosing the tail *after* looking (testing $\theta > 1/2$ because Karim happens to lead) silently converts a two-sided question into a one-sided test and doubles the real level: the freedom to pick the favorable tail is worth a factor of two. Section 3.5 meets the industrial-scale version of this mistake.

3.3.2 Skewed nulls

The doubling in the previous computation used the symmetry of $\mathrm{Bin}(20, 1/2)$, and symmetry is the exception, not the rule. Karim, on a roll, claims he converts 80% of his penalty kicks. A month of practice data: 16 scored out of 25. The null is now $H_0 : \theta = 0.8$ against $H_1 : \theta \neq 0.8$,

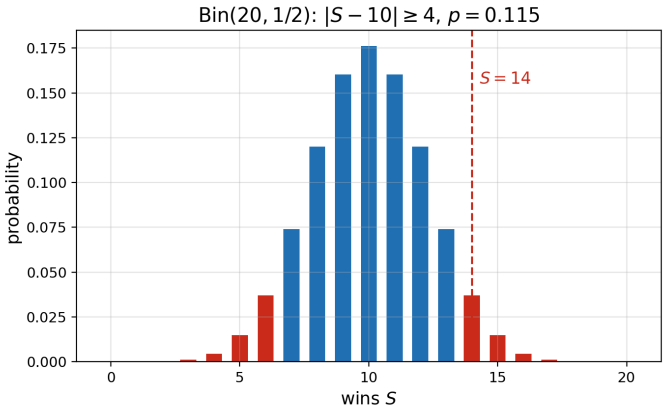


Figure 3.4: Under the symmetric null $\mathrm{Bin}(20, 1/2)$, the two-sided region $|S - 10| \geq 4$ has two mirror-image tails, and the two-sided p-value doubles the one-sided one.

and the null distribution $\mathrm{Bin}(25, 0.8)$ is visibly skewed (figure 3.5): the two tails are no longer mirror images, and "as extreme on the other side" has no unique meaning. The observed left tail is $\mathbb{P}(S \leq 16) = 0.047$. Two conventions coexist for the two-sided p-value: *double the smaller tail* ($p = 0.094$), or *sum all outcomes at most as likely as the observed one* ($p = 0.074$ here). They disagree, and neither is wrong; doubling is common in textbooks, while the exact binomial tests of scipy and R (`binomtest`, `binom.test`) use the second; honest practice states which one is used. The lesson survives beyond binomials: with an asymmetric null distribution, two-sided p-values involve a convention, and the symmetric case of section 3.3.1 was a coincidence of $\theta_0 = 1/2$.

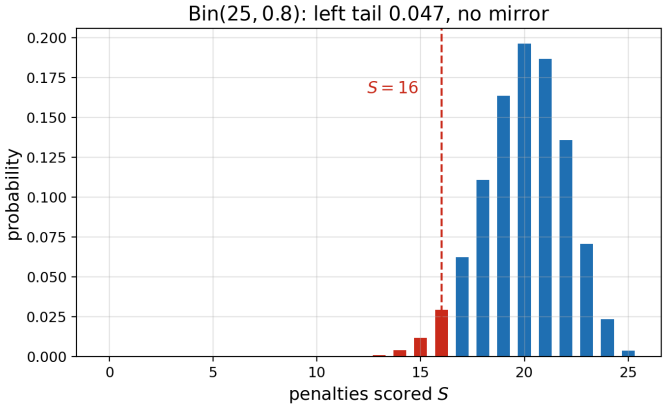


Figure 3.5: The penalty null $\mathrm{Bin}(25, 0.8)$ is skewed: the observed left tail (0.047) has no mirror image, and the two-sided p-value depends on a convention.

3.3.3 Composite nulls

Section 3.2.5 stated the dispute in its full form, $H_0 : \theta \leq 1/2$, a *composite* null: a whole range of parameter values with no single distribution to compute under. It asserted without proof that the rejection probability is largest at the boundary, which is what reduces the composite null to its boundary. Here is the statement.

Property 3.3.1 (Composite one-sided nulls reduce to the boundary)

For every k, the tail probability $\theta \mapsto \mathbb{P}_\theta(S \geq k)$ is nondecreasing in θ. Consequently

$$\sup_{\theta \leq 1/2} \mathbb{P}_\theta(S \geq k) = \mathbb{P}_{1/2}(S \geq k),$$

so a test of the form "reject when $S \geq k$" whose false-rejection probability is at most α at $\theta = 1/2$ has size at most α over the whole composite null $\theta \leq 1/2$.

The monotonicity is intuitive: a better player produces more wins. The smaller θ is, the less often it produces $S \geq k$, so among all the values in the null, $\theta = 1/2$ (the largest, the one closest to the alternative) is the *least favorable* null value, the one where a false rejection is most likely: a false-rejection probability held below α at $\theta = 1/2$ is held below α at every smaller value automatically, so controlling it controls the whole null. Testing $\theta \leq 1/2$ against $\theta > 1/2$ therefore uses the same statistic, the same quantile and the same p-value as testing $\theta = 1/2$ against $\theta > 1/2$. (The proof of the monotonicity is a short coupling argument.) This reduction is general for one-sided problems with this monotonicity, and it is why one-sided tests of a regression coefficient, of the kind met in first year ($\beta \leq 0$ against $\beta > 0$), are computed at the single point $\beta = 0$.

3.4 Statistical significance is not statistical importance

Karim takes the dispute online and grinds a full ladder season: $n = 2000$ ranked games, a 53% win rate, that is $S = 1060$ wins. The one-sided test of $H_0 : \theta = 1/2$ now rejects emphatically: the binomial tail gives $p = \mathbb{P}(S \geq 1060) = 0.0039$. Highly significant. Has Karim shown he is *much* better? The estimate answers a different question than the test, and both deserve to be reported. The point estimate is $\hat{\theta} = 0.53$, and the 95% confidence interval from the central limit theorem of chapter 1 with a plug-in standard error, $\hat{\theta} \pm 1.96\sqrt{\hat{\theta}(1-\hat{\theta})/n}$, is $[0.508, 0.552]$: an edge of about one to five points (at the midpoint, one extra win every 33 games). The rejection is decisive; the effect is small.

Key idea

Significance is not size. The p-value measures the incompatibility of the data with $\theta = 1/2$, in the direction specified by the alternative; it does not measure the size of the effect. With n large enough, *every* nonzero effect becomes significant: at $n = 2000$, a 53% player rejects fair play with room to spare. Report the estimate and its confidence interval next to any p-value: the p-value alone is a claim about the null value, not about magnitude.

Tests and confidence intervals are, in fact, the same object seen from two sides. The 95% interval $[0.508, 0.552]$ is exactly the set of values θ_0 that the level-5% two-sided Wald test (the statistic $(\hat{\theta} - \theta_0)/\widehat{\mathrm{se}}$, with the same estimated standard error) would *not* reject on these data (other tests, such as the exact binomial one, give nearly the same set): $1/2$ falls outside the interval, and the two-sided test of $\theta = 1/2$ rejects; 0.53 falls inside, and the test of $\theta = 0.53$ does not. Reading an interval as "all the nulls these data leave standing" often communicates both significance and size at once, which is why reporting the interval is never a downgrade from reporting the test.

3.5 Multiple testing

Every number in this chapter was computed under assumptions, most of them loaded as quietly as Arbuthnot's: independent games, a constant θ and, everywhere, the reading of *one* test at level α. Step 7 of the ladder says to stress-test such assumptions before trusting a verdict. This

section works out the stress test for the last one, the assumption broken most often in practice: what survives of a 5% level when two hundred tests are read at once.

3.5.1 The smurf screen

The platform hosting the ladder has a smurfing problem: experienced players opening fresh accounts to stomp beginners. The anti-cheat team screens $m = 200$ accounts, testing each one's win rate against $H_0 : \theta = 1/2$ over its last 60 games, at level 5% each. Every individual test is calibrated. The screen as a whole is not. If all 200 accounts are honest, the expected number of false flags is at most $200 \times 0.05 = 10$ (about 9 for this discrete test, which rejects at $S \geq 37$ with real size 0.046), and for independent tests the probability of at least one false flag is

$$\mathbb{P}(\text{at least one false rejection}) = 1 - (1 - \alpha)^m, \tag{37}$$

already 40% at $m = 10$ and 0.99996 at $m = 200$ (figure 3.6). The quantity on the left is called the *family-wise error rate* (FWER). A per-test level does not control it: with enough calibrated tests, at least one false discovery becomes almost certain. This is the large-scale version of the tail-picking of section 3.3.1: many looks at noise, each calibrated on its own, together very likely to flag something.

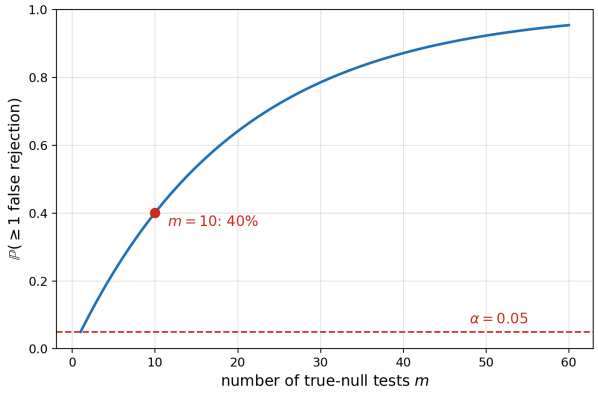


Figure 3.6: The probability (37) of at least one false rejection among m independent true-null tests at $\alpha = 5\%$: 40% at $m = 10$, near certainty long before $m = 200$.

3.5.2 Controlling the family-wise error

The oldest fix is *Bonferroni*: test each account at level α/m instead of α. By the union bound, which requires no independence, the probability that *any* true null is rejected is at most $m \cdot \alpha/m = \alpha$. The price is severity: at $m = 200$ the working threshold is $0.05/200 = 0.00025$, and moderate effects are lost. *Holm's procedure* spends the same budget more efficiently (sort the p-values, compare the smallest to α/m, the next to $\alpha/(m-1)$, and so on until the first failure) and rejects everything Bonferroni rejects, sometimes more, with the same FWER guarantee (stated here without proof).

3.5.3 Controlling the false discovery rate

For a screening problem like the smurf hunt, guarding against any single false flag is often too strict; what matters is that the flagged *list* be mostly right. The *false discovery rate* (FDR) is the expected proportion of false flags among the flags, and the *Benjamini–Hochberg* (BH) procedure controls it at level α [3]: sort the p-values $p_{(1)} \leq \cdots \leq p_{(m)}$, find the largest k such that

$$p_{(k)} \;\leq\; \frac{k}{m}\,\alpha, \tag{38}$$

and flag the k smallest. The threshold is a rising line rather than a floor: the more genuinely small p-values the screen contains, the more the procedure lets through. The BH theorem covers independent tests and extends to certain positively dependent families, not to arbitrary dependence; Bonferroni and Holm, resting on the union bound alone, need no such qualification.

The simulated screen makes the comparison concrete. Among the 200 accounts, 195 are honest and 5 are planted smurfs with a true 80% win rate. Naive per-account testing at 5% flags 20 accounts: the 5 smurfs and 15 innocent players (on the high side, since about 9 false flags are expected among 195 honest accounts, and 15 or more occur with probability about 0.04). Bonferroni flags exactly the 5 smurfs, and so does BH (figure 3.7); with weaker smurfs BH would typically keep more of the real ones than Bonferroni, at the cost of a controlled fraction of false flags.

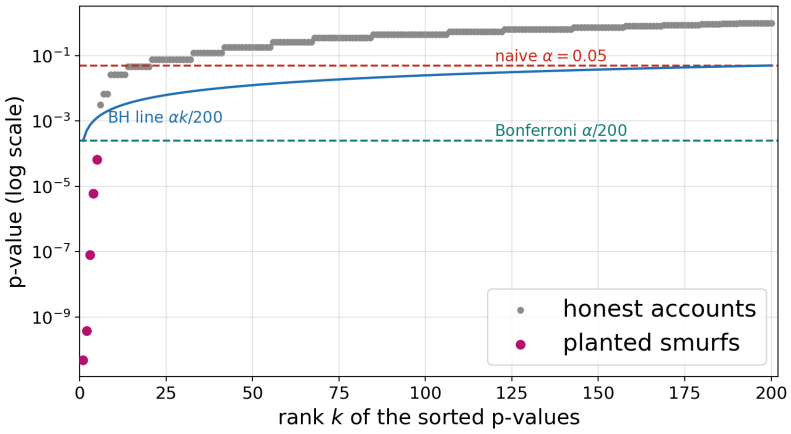


Figure 3.7: The 200 sorted p-values of the smurf screen (log scale). The five planted smurfs sit far below the Bonferroni threshold $\alpha/200$; fifteen honest accounts fall between the BH line and the naive 5% line and are flagged only by the naive screen. The staircase pattern in the honest p-values is the discreteness of section 3.2.4 again: with 60 games, only 61 p-values are possible.

Reality check

Which correction to use is a question about the purpose of the analysis, because the two guarantees differ in strictness. FWER control is the conservative one: it bounds the probability of even a single false flag anywhere in the family, and pays for that with fewer discoveries. FDR control is more permissive: it flags more, and guarantees only that the flagged list is, on average, no more than a fraction α false flags. A confirmatory analysis, whose conclusion will support a published claim or a costly decision, controls the FWER. An exploratory screen over many candidates controls the FDR, accepting that expected fraction of false leads among its discoveries in exchange for more of them. In both cases the correction is chosen *before* the p-values are examined, for the same reason the level and the alternative are: a procedure adjusted after seeing what it saves no longer controls any error rate.

3.6 Conclusion

The chapter's product is the testing ladder boxed in section 3.2.5, now climbed in full:

step	in this chapter
1. the question	is Karim better, or lucky?
2. H_0 vs H_1	$\theta \leq 1/2$ against $\theta > 1/2$; the alternative decided every tail
3. the statistic	the number of wins S
4. null distribution	$\mathrm{Bin}(20, 1/2)$, exactly or by simulation
5. the p-value	0.058: a gray zone, not a verdict
6. effect size	2000 games: $p = 0.0039$ yet $\hat{\theta} \in [0.508, 0.552]$: significant and small
7. stress tests	200 honest accounts at 5%: false flags almost certain; Bonferroni/Holm or BH repair
8. conclude in context	evidence under assumptions, never an automatic verdict

Failing to reject the null means insufficient evidence against it, never evidence that it is true: twenty games could not distinguish equal skill from a 70% edge. The alternative, chosen from the question before the data, decides the rejection region, the quantile, and the p-value. Significance is not size: enough games make a trivial edge arbitrarily significant. And every level ever quoted rests on assumptions (independence, distributional shape, a single test) that can fail as silently as Arbuthnot's fair coin was assumed; his overreach, reading a cause into a rejection, is the standing warning of step 8.

The next chapter puts this ladder to work on a real model: a regression of customer spending on its covariates, with a t-test for every coefficient and a screen of twenty-eight coefficients at once, the multiple-testing problem of section 3.5 in a regression.

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[2] J. Neyman and E. S. Pearson. On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society A*, 231:289–337, 1933.

[3] Y. Benjamini and Y. Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1):289–300, 1995.

Chapter 4
Regression as statistical modeling

For two chapters we modeled the *joint* distribution of the row: one Gaussian when the data form one cluster, a mixture of Gaussians when they form several. That is the most complete thing a statistician can do with a table, and it is not always the right thing. It can be *too complicated*: the joint model's number of parameters grows quadratically with the dimension, and a mixture multiplies that count by the number of components. And it can be *unnecessary*: often one component of the vector has a special status (the quantity to be predicted, priced, or acted on), and the question is not how the whole row is distributed but what that one component is worth when the others are known. Call it Y, and call the rest $\mathbf{X}$. The prologue already named the way out: the joint distribution factors into the law of $\mathbf{X}$ times the conditional law of Y given $\mathbf{X}$, and prediction needs only the second factor. This chapter models that factor alone, regardless of what the distribution of $\mathbf{X}$ looks like, and it does so in the spirit of Part 1: a parametric model, its assumptions examined one at a time, and exact inference. The model is one every reader has met before, the linear model of first year; the point of the chapter is to re-read it as a *statistical model*: what each assumption buys, what the tests actually say, and what happens to twenty-eight of them at once.

Throughout, the course conventions hold: bold letters denote vectors and matrices, plain italic letters denote scalars.

4.1 From the joint to a conditional model

Return to the store of chapter 2, and zoom in on the segment the mixture called the *regulars*: $n = 400$ customers. For each of them the database records the monthly spend, and three more columns: visits per month, household size, and distance to the store in kilometers,

	spend (Dh)	visits / month	household	distance (km)
customer 1	362	13.0	3	7.8
customer 2	222	10.7	2	6.6
customer 3	977	11.7	5	4.9
customer 4	753	13.8	4	4.2
⋮	⋮	⋮	⋮	⋮
customer 400	334	11.4	2	6.2

The store's question is not symmetric in these four quantities. Spend is the one it acts on (revenue to forecast, offers to target), and the other three are context: how much does an extra visit bring? does distance cost money? what should the store expect from a new customer whose three covariates are known? Write Y for the spend and $\mathbf{x} = (x_1, \dots, x_p)^\top$ for the covariates ($p = 3$ here).

A full chapter-1 treatment would model the joint distribution of the vector $(\mathbf{X}, Y)$ on $\mathbb{R}^{p+1}$. A Gaussian model of the four-dimensional row needs $(p+1)+(p+1)(p+2)/2$ parameters, asserts joint Gaussianity of covariates that may be nothing of the sort (household size takes whole values; distance is skewed), and spends most of its parameters describing how the covariates co-vary among themselves, which the store's question never asks. The conditional route keeps only what the question needs. When a vector is jointly Gaussian, the conditional law of one coordinate

given the others is exactly Gaussian, with a mean *affine* in the observed values and a variance that does not depend on them. The conditional model takes that conclusion and promotes it to an assumption, while dropping every claim about $\mathbf{X}$:

Definition 4.1.1 (Gaussian linear model)

The *Gaussian linear model* for a response Y given covariates $\mathbf{x} \in \mathbb{R}^p$ is the conditional model

$$Y \mid \mathbf{X} = \mathbf{x} \;\sim\; \mathcal{N}(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x},\; \sigma^2), \qquad \text{equivalently} \qquad Y = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x} + \varepsilon, \tag{39}$$

with parameters $\beta_0 \in \mathbb{R}$, $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\sigma^2 > 0$, and with $\varepsilon := Y - \beta_0 - \boldsymbol{\beta}^\top \mathbf{X}$ the deviation from the affine mean. The assumptions, stated separately:

(A1) *correct mean (linearity and exogeneity)*: $\mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}) = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}$ for every $\mathbf{x}$, equivalently $\mathbb{E}(\varepsilon \mid \mathbf{X}) = 0$;

(A2) *independent rows*: the customers are sampled independently (given the design, the deviations $\varepsilon_1, \dots, \varepsilon_n$ of distinct rows are independent);

(A3) *homoscedasticity*: $\mathbb{V}(\varepsilon \mid \mathbf{X}) = \sigma^2$, the same constant at every $\mathbf{x}$;

(A4) *conditional Gaussianity*: given $\mathbf{X} = \mathbf{x}$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.

A word on (A1), which some textbooks split into two: "linearity" of the mean and "exogeneity" of the noise. With ε *defined* as the deviation from the affine mean, the two are one and the same statement, and we state it once. The content that matters in practice is this: (A1) claims that the conditional mean is correctly specified *with the covariates included in the model*. Its typical failure is not a curve where a line was drawn but an *absent* covariate: when a variable that influences Y is left out and is correlated with an included one, the included coefficient absorbs part of the missing effect. Section 4.5 will show why this particular failure is the hardest to detect.

Nothing is assumed about the distribution of $\mathbf{X}$: the covariates may be discrete, skewed or dependent. This is where the economy comes from: instead of a distribution on $\mathbb{R}^{p+1}$, the model has $p + 2$ parameters. Figure 4.1 draws the model's claim on the regulars' data, for the single covariate visits: one straight line for the conditional mean, and around it the same Gaussian at every visit frequency (same shape, same width, only the center moving along the line). Whether the data agree with that claim is a question the chapter takes up in section 4.5; until then, they are working assumptions.

Reality check

(A1)–(A4) are not hypotheses to be verified and ticked off: no dataset can prove that a conditional mean is exactly affine or that errors are exactly Gaussian. They are working assumptions. Each one, as section 4.3 maps out, buys a specific inferential product, and each one, when wrong, silently invalidates what it bought. The honest attitude is the one this course keeps returning to: put the assumptions to work, then gather evidence of how far off they are (section 4.5), knowing the evidence can refute them but never certify them.

4.2 Least squares is maximum likelihood

Stack the n observed rows: let $\mathbf{y} = (y_1, \dots, y_n)^\top$ collect the responses and let the *design matrix* $\mathbb{X} \in \mathbb{R}^{n \times (p+1)}$ have rows $(1, \mathbf{x}_i^\top)$ (a column of ones for the intercept, then the covariates). Absorbing β_0 into $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)^\top$, the n equations (39) become

$$\mathbf{y} = \mathbb{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 \mathrm{I}_n), \tag{40}$$

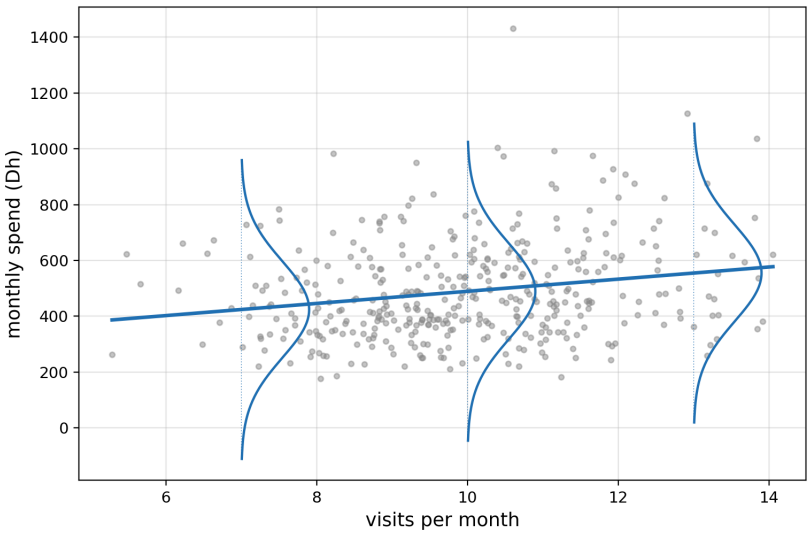


Figure 4.1: The model's claim, drawn on the regulars' spend against visits: a straight conditional mean and the same Gaussian conditional distribution at every visit frequency (three slices shown). Whether the data agree is section 4.5's question.

the independence across rows being assumption (A2) at work. Under the model, the likelihood of the observed responses given the design is

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\sigma^2}\right) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|\mathbf{y} - \mathbb{X}\boldsymbol{\beta}\|^2}{2\sigma^2}\right),$$

writing $\mathbf{x}_i$ for the i-th row of the design. Whatever σ, maximizing L over $\boldsymbol{\beta}$ means minimizing the exponent: maximum likelihood in the Gaussian linear model *is* least squares, $\min_\beta \|\mathbf{y} - \mathbb{X}\boldsymbol{\beta}\|^2$, the first-year criterion, now obtained as a consequence of the model.

Theorem 4.2.1 (OLS estimator)

If $\mathbb{X}^\top\mathbb{X}$ is invertible, the likelihood is maximized at

$$\hat{\boldsymbol{\beta}} = (\mathbb{X}^\top\mathbb{X})^{-1}\mathbb{X}^\top\mathbf{y}, \qquad \hat{\sigma}^2_{\text{MLE}} = \frac{\text{RSS}}{n}, \quad \text{RSS} = \|\mathbf{y} - \mathbb{X}\hat{\boldsymbol{\beta}}\|^2. \tag{41}$$

Proof. The map $g(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbb{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top\mathbf{y} - 2\boldsymbol{\beta}^\top\mathbb{X}^\top\mathbf{y} + \boldsymbol{\beta}^\top\mathbb{X}^\top\mathbb{X}\boldsymbol{\beta}$ has gradient $\nabla g = -2\mathbb{X}^\top\mathbf{y} + 2\mathbb{X}^\top\mathbb{X}\boldsymbol{\beta}$. Setting it to zero gives the *normal equations* $\mathbb{X}^\top\mathbb{X}\,\boldsymbol{\beta} = \mathbb{X}^\top\mathbf{y}$, whose unique solution is $\hat{\boldsymbol{\beta}}$ when $\mathbb{X}^\top\mathbb{X}$ is invertible; the critical point is the global minimum because the Hessian $2\mathbb{X}^\top\mathbb{X}$ is positive definite (for any $\mathbf{u} \neq 0$, $\mathbf{u}^\top\mathbb{X}^\top\mathbb{X}\mathbf{u} = \|\mathbb{X}\mathbf{u}\|^2 > 0$ by invertibility). With $\hat{\boldsymbol{\beta}}$ fixed, the likelihood as a function of σ^2 is $(\sigma^2)^{-n/2}e^{-\text{RSS}/(2\sigma^2)}$, maximized at $\sigma^2 = \text{RSS}/n$, the same one-dimensional maximization as chapter 1's Gaussian MLE. □

As in chapter 1, the MLE of the variance is biased; the unbiased version divides by the residual degrees of freedom,

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n - p - 1}, \tag{42}$$

$n - p - 1$ playing the role that $n - 1$ played for the sample variance: one degree of freedom is spent per fitted coefficient.

The estimator fluctuates, with a known law. Substituting (40) into (41),

$$\hat{\boldsymbol{\beta}} = (\mathbb{X}^\top \mathbb{X})^{-1}\mathbb{X}^\top(\mathbb{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbb{X}^\top \mathbb{X})^{-1}\mathbb{X}^\top \boldsymbol{\varepsilon}\,:$$

conditionally on the design, $\hat{\boldsymbol{\beta}}$ is an *affine map of a Gaussian vector*, precisely the situation chapter 1's stability property handles. Its mean is $\boldsymbol{\beta}$ and its covariance is $\mathbf{A}\,\sigma^2 \mathrm{I}_n\,\mathbf{A}^\top$ with $\mathbf{A} = (\mathbb{X}^\top \mathbb{X})^{-1}\mathbb{X}^\top$, which collapses to:

$$\hat{\boldsymbol{\beta}} \mid \mathbb{X} \;\sim\; \mathcal{N}(\boldsymbol{\beta},\; \sigma^2(\mathbb{X}^\top \mathbb{X})^{-1}). \tag{43}$$

Moreover $\mathrm{RSS}/\sigma^2 \sim \chi^2(n-p-1)$ and RSS is independent of $\hat{\boldsymbol{\beta}}$ (the Cochran theorem again, stated without proof exactly as in chapter 1). These two facts are the basis of all the inference in this chapter.

4.3 What each assumption buys

The definition listed four assumptions. They are not a package: each one buys a specific product, and the products stack. Adding them one at a time, starting from none, shows what each contributes.

No assumptions: a fit. With no model at all, OLS is arithmetic: the line (or hyperplane) closest to the cloud in squared distance. It always exists and always produces numbers, but nothing can be said about those numbers: they estimate no defined target and carry no error bar. Inference requires a model.

A correct mean buys unbiasedness. Under (A1), the estimator has a target and hits it on average: taking conditional expectations in the displayed decomposition above,

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid \mathbb{X}) = \boldsymbol{\beta} + (\mathbb{X}^\top \mathbb{X})^{-1}\mathbb{X}^\top\, \mathbb{E}(\boldsymbol{\varepsilon} \mid \mathbb{X}) = \boldsymbol{\beta},$$

the last step being exactly (A1)'s exogeneity form. This is the assumption that most often fails in practice: if a covariate that influences Y is missing from the model and correlated with an included one, the noise absorbs it, (A1) fails, and the included coefficient is biased, partly reporting the absent variable's effect. No amount of data repairs the bias.

Independence buys the error accounting. Assumption (A2) is easy to forget because the customers of a loyalty file satisfy it so naturally, and it is the one that makes the covariance computation of section 4.2 possible at all: independent rows mean the error covariance matrix is diagonal, so information adds up row by row. When rows are dependent (several customers from the same household, or repeated measurements of the same person), the file contains fewer effective observations than lines, and the standard errors computed as if n rows were n pieces of information come out wrong, usually too small. This failure is in one respect worse than the funnel of section 4.5: neither residual plot shows it, because each residual taken alone looks perfectly ordinary.

Homoscedasticity buys the standard errors. Under (A1)–(A3), the covariance formula in (43) is correct, so the reported standard error of each coefficient, $\mathrm{SE}(\hat{\beta}_j) = \hat{\sigma}\sqrt{[(\mathbb{X}^\top \mathbb{X})^{-1}]_{jj}}$, estimates the real sampling fluctuation. If the noise variance instead changes with $\mathbf{x}$, OLS remains unbiased but the formula is wrong, and every quantity computed from a standard error (tests, intervals) is wrong with it. The same assumptions also carry a classical optimality: among all linear unbiased estimators, OLS has the smallest variance (the Gauss–Markov theorem, a classical result we use without proof).

Gaussianity buys exact finite-n inference. Under the full model (A1)–(A4), the laws of section 4.2 are exact for every n, and they deliver the t and F tests and confidence intervals of the next section at their stated levels, for $n = 30$ as for $n = 400$, with no large-sample approximation. Without (A4), those levels are only approximations, good when n is large (the CLT at work), unreliable when it is not.

Each of these assumptions can be wrong, and its inferential product is then lost. The next section puts the products to work; section 4.5 then confronts the assumptions with the data, and the chapter's closing table lists each assumption, its product, and whether the diagnostics can detect its failure.

4.4 Testing a coefficient

Does distance to the store really cost money, once visits and household size are accounted for? The question is about β_{dist}, and the model makes it answerable with a number.

4.4.1 Testing one coefficient

From (43), the j-th coefficient satisfies $\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2 v_j)$ with $v_j = [(\mathbb{X}^\top \mathbb{X})^{-1}]_{jj}$. Standardizing and replacing the unknown σ by $\hat{\sigma}$ produces a statistic whose law is exactly known under the model; the replacement is legitimate because RSS is independent of $\hat{\boldsymbol{\beta}}$:

$$T_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{v_j}} \;\sim\; t(n-p-1), \tag{44}$$

Student's distribution with $n - p - 1$ degrees of freedom. Two products follow immediately. A *confidence interval* at level $1-\alpha$: $\hat{\beta}_j \pm t_{1-\alpha/2}(n-p-1)\,\mathrm{SE}(\hat{\beta}_j)$. And a *test* of $H_0 : \beta_j = 0$: reject when the observed $|t_j| = |\hat{\beta}_j|/\mathrm{SE}(\hat{\beta}_j)$ is beyond the quantile. The *p-value* attached to the test is the probability, *computed under* H_0, of drawing a statistic at least as extreme as the one observed: $p_j = \mathbb{P}_{H_0}(|T_j| \geq |t_j|)$.

On the regulars ($n = 400$, $p = 3$, hence 396 degrees of freedom), the fitted spend equation gives $\hat{\beta}_{\text{visits}} = 24.6$ Dh per visit with SE $= 3.8$, hence $t = 6.5$ and $p = 2 \cdot 10^{-10}$; and $\hat{\beta}_{\text{dist}} = -16.6$ Dh per kilometer with SE $= 3.4$, $t = -4.9$, $p = 1.6 \cdot 10^{-6}$. Under the model, distance costs money: about seventeen dirhams of monthly spend per kilometer, with an uncertainty the interval makes precise. (These numbers hold under the model; section 4.5 examines whether the model itself holds.)

What the numbers do not say. The t-test is widely used and often misread. The following misconceptions are common.

- *A large p-value does not mean H_0 is true.* Failing to reject is absence of evidence, not evidence of absence: the same $p = 0.4$ arises for a null effect and for a real effect measured with too little data. One never "accepts" H_0.
- *The p-value is not $\mathbb{P}(H_0 \mid \text{data})$.* It is computed assuming H_0; it quantifies the surprise of the data under that assumption, not the probability of the assumption.
- *Significance is not importance.* The t-statistic scales with $\sqrt{n}$: with enough rows, a coefficient of one centime per kilometer becomes "highly significant". Read the estimate and its confidence interval (quantities in dirhams), not the stars next to the p-value.
- *A large R^2 is neither necessary nor sufficient.* Valid inference on coefficients does not require a large R^2, and a large R^2 certifies nothing about the assumptions the inference rests on. The paragraph after this list unpacks both directions.

- *Coefficients are ceteris paribus, and local.* $\hat{\beta}_{\text{visits}}$ is the effect of one more visit *holding household size and distance fixed*, estimated over the observed range of the data; the fitted equation says nothing about a customer visiting forty times a month. Extrapolation is silent: the formula produces a number without warning that it has left the data.

What R^2 does and does not measure. R^2 measures one thing: the share of the variance of Y that the fitted mean absorbs, or equivalently, how small the noise σ^2 is relative to the spread of the response. The validity of inference does not depend on that share, in either direction. *Not necessary*: nothing in the construction of (44) involves R^2; the exact laws behind the t-statistic require (A1)–(A4), not a small σ. Under a valid model with a noisy response, a low R^2 means individual predictions are imprecise, and the intervals say so honestly by being wide; but their coverage, and the level of every test, remain exact. A small real effect in a noisy response is estimated *correctly*, just less precisely. *Not sufficient*: R^2 is computed from the fit alone and never looks at the assumptions, so it cannot certify them. The chapter's own fits make this concrete: the raw-spend regression of section 4.5 reports a respectable $R^2 = 0.49$ while the funnel beneath it invalidates every standard error; and adding the redundant items column of the multicollinearity paragraph below leaves that same R^2 unchanged (0.4908 to 0.4909) while every t-test on the pair dissolves.

Reality check

This section has said "the effect of one more visit" and "distance costs money". Under (A1)–(A3) such phrases describe an *association* in the population as it is: among the store's customers, those who live one kilometer farther spend less on average, the other included covariates held fixed. Nothing in the model says what would happen *if* the store induced an extra visit or relocated a customer: a causal claim, which requires that no unrecorded variable drives both the covariate and the response. No assumption of this chapter provides that, and an omitted variable can manufacture a convincing "effect" out of nothing. Reading coefficients as causal effects is a decision that must be defended outside the model.

4.4.2 Predicting one customer

The same machinery gives the uncertainty of an individual prediction. For a new customer with covariates $\mathbf{x}_0$ (as a design row, with the leading 1), the point prediction is $\hat{y}_0 = \mathbf{x}_0^\top \hat{\beta}$, and its error has two independent sources: the estimation error of $\hat{\boldsymbol{\beta}}$, with variance $\sigma^2\, \mathbf{x}_0^\top (\mathbb{X}^\top \mathbb{X})^{-1} \mathbf{x}_0$ by (43), and the customer's own noise ε_0, with variance σ^2. Adding them and studentizing as in (44) gives:

Property 4.4.1 (Prediction interval)

Under (A1)–(A4), a level-$(1-\alpha)$ prediction interval for the response of a new, independent row $\mathbf{x}_0$ is

$$\hat{y}_0 \;\pm\; t_{1-\alpha/2}(n-p-1)\, \hat{\sigma} \sqrt{1 + \mathbf{x}_0^\top (\mathbb{X}^\top \mathbb{X})^{-1} \mathbf{x}_0}\,, \qquad \hat{y}_0 = \mathbf{x}_0^\top \hat{\beta}. \tag{45}$$

The two terms under the root continue the R^2 discussion. The design term $\mathbf{x}_0^\top (\mathbb{X}^\top \mathbb{X})^{-1} \mathbf{x}_0$ shrinks with n (for a typical customer of the log-spend model it is 0.003); the constant 1 does not. For any reasonable sample size the interval's width is essentially $2\, t_{1-\alpha/2}\, \hat{\sigma}$, governed by the noise level, which is exactly what R^2 measures the size of. On the regulars, a customer with 10 visits, a household of 3 and 5 km gets a median spend prediction of 435 Dh with a 95% interval of $[268, 707]$ Dh, obtained by exponentiating the log-scale interval 6.08 ± 0.49. (Monotone transformations carry quantiles, so the interval survives the return to dirhams.) The interval is honest and wide: with $R^2 = 0.51$, half the variance of log spend is noise, and no amount of data narrows an individual prediction below it. The R^2 that was irrelevant to coefficient inference is decisive for individual prediction.

4.4.3 Several coefficients at once

The F test. To test a group of coefficients jointly (do the store's three covariates explain anything at all?), one compares the residual sums of squares of the model with and without them: with q coefficients set to zero under H_0,

$$F = \frac{(\mathrm{RSS}_0 - \mathrm{RSS})/q}{\mathrm{RSS}/(n-p-1)} \;\sim\; F(q,\, n-p-1) \quad \text{under } H_0.$$

A large F means the q extra coefficients reduce the residual sum of squares by more than chance would explain: one joint answer, where the t-tests give q marginal ones. The distinction matters in practice; the following situation, common in any database with redundant columns, shows why.

When covariates overlap: multicollinearity. Suppose the store also records the number of *items* bought per month. Regulars buy roughly the same basket per trip, so items is very nearly a multiple of visits: in our data the correlation between the two columns is $\rho = 0.988$. Adding items to the regression of the worked example above degrades the inference on visits sharply: its standard error jumps from 3.8 to 24.3 (more than six times larger), its t-statistic drops from 6.5 to 1.3 ($p = 0.20$), and items itself comes out at $p = 0.79$. Read one t-test at a time, $n = 400$ customers cannot confirm that either of the two matters. The F test of the pair, however, shows clearly that together they do: $F = 21.2$, $p = 2 \cdot 10^{-9}$.

There is no contradiction: the two tests answer different questions. Each t-test asks whether its covariate still matters *once the other is in the model*; and given items, visits adds almost nothing (nor the reverse). The F test asks whether the pair matters at all, and it plainly does. The mechanism sits in the formula the chapter has used throughout: two nearly proportional columns make $\mathbb{X}^\top\mathbb{X}$ nearly singular, and the diagonal entries of its inverse, the v_j of (44), blow up. This is the same near-singularity chapter 1 met in covariance estimation. For two standardized covariates with correlation ρ, the standard errors inflate by the factor $1/\sqrt{1-\rho^2}$, which at the observed $\rho = 0.988$ equals 6.4, matching the jump from 3.8 to 24.3 (the residual $\hat{\sigma}$ barely moves when items is added, so the factor carries through). This is *multicollinearity*, and it dictates how the output may be read: the fitted predictions are unharmed (the model knows what spend to expect, since either column carries the information), but the *attribution* of the effect between near-twin covariates is undetermined by the data, and t-tests must not be read as a ranking of variable importance. What to do about it, beyond the obvious housekeeping of dropping a redundant column, is a question the next chapter takes up.

4.5 Diagnostics: evidence, not proof

4.5.1 Residual plots and the log repair

Everything so far holds *under the model*. The reality check of section 4.1 demands the confrontation: the assumptions cannot be verified, but the fitted model leaves traces (the residuals), and residuals can contradict it. Two plots do most of the work.

- *Residuals against fitted values.* Under (A1)–(A3) the residuals should hover around zero in a band of constant width, at every fitted level. Curvature indicates a conditional mean that is not affine in the included covariates, the visible part of (A1); a funnel (spread growing with the fitted value) indicates heteroscedasticity (A3 violated).
- *QQ-plot.* Sorted standardized residuals against Gaussian quantiles: systematic departure from the diagonal (curved tails, an S shape) indicates non-Gaussian errors (A4 violated).

Figure 4.2 (top row) runs both on the regulars' regression of section 4.4, and both plots contradict the model. The residual plot funnels clearly: the spread at a fitted spend of 700 Dh

is more than half again the spread at 350 Dh, so (A3) fails. With it fail the standard errors, t-statistics and p-values just computed, all of which rest on a constant σ. The QQ-plot bends away from the diagonal in the right tail: big spenders are more extreme than a Gaussian allows.

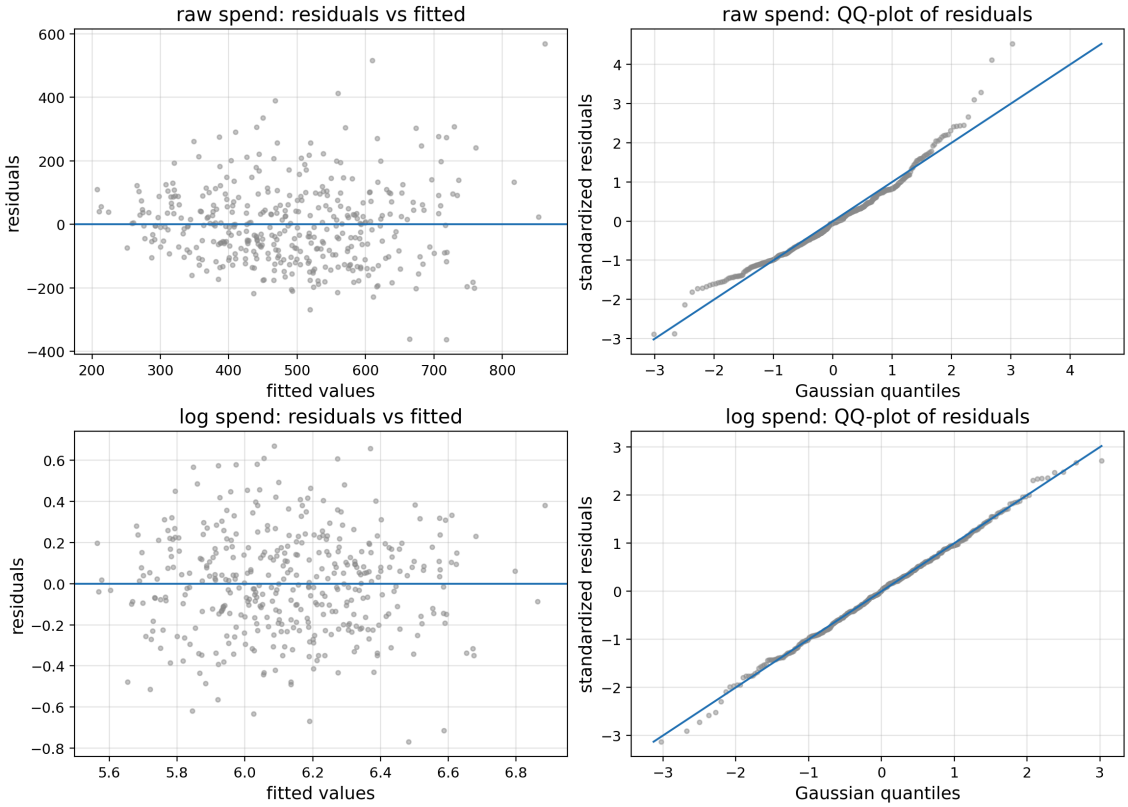


Figure 4.2: Diagnostics for the regulars' regression. Top: raw spend; the residuals funnel outward (A3 fails) and the QQ-plot bends in the right tail (A4 strained). Bottom: the same customers, response log(spend); both plots are clean. The remedy was a modeling decision, not a test.

The remedy here is a modeling decision rather than an adjustment to the tests. Nothing in the chapter requires the modeled response to be the raw recorded quantity: for any strictly increasing transformation g, one may posit the Gaussian linear model of definition 4.1.1 for the *transformed* response $g(Y)$,

$$g(Y) \mid \mathbf{X} = \mathbf{x} \;\sim\; \mathcal{N}(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x},\; \sigma^2), \tag{46}$$

and every result of sections 4.2–4.4 applies verbatim with y_i replaced by $g(y_i)$. The choice of g is part of the model specification, on the same footing as (A1)–(A4), and it is judged by the same diagnostics; the identity $g(y) = y$ (modeling raw spend) was itself such a choice, and the funnel is its refutation. Here the nature of the response points to $g = \log$: spend is positive, and effects on it plausibly act multiplicatively (an extra household member scales a basket rather than adding a fixed amount). The adopted model is therefore (46) with $g = \log$: $\log Y \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}, \sigma^2)$. The bottom row of figure 4.2 refits the same three covariates on the log response: the funnel is gone, the QQ-plot straightens, and the model's assumptions are no longer contradicted by its own residuals. The coefficients change meaning accordingly: on the log scale each unit of a covariate multiplies expected spend by a fixed factor: one more monthly visit by $e^{0.048} \approx 1.05$ (five percent), one more household member by $e^{0.17} \approx 1.19$, one more kilometer by $e^{-0.033} \approx 0.97$. These are readings at least as useful to the store as dirham amounts. One caution when mapping predictions back to dirhams: $e^{\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}}$ is the conditional *median* of spend, not its mean; for a log-Gaussian response the mean is larger, by the factor $e^{\sigma^2/2}$.

4.5.2 When the variance will not be repaired

A transformation fixed (A3) here because the effects were themselves multiplicative; nothing guarantees that a scale exists on which the mean is affine *and* the variance is constant. When heteroscedasticity remains, a standard partial remedy keeps the OLS estimates, still unbiased under (A1), and replaces the variance formula, which is what the funnel falsifies, by an estimate built from the squared residuals themselves, valid without a common σ^2: the *heteroscedasticity-consistent* (robust, or White) standard errors, which we name as a tool and use without proof. They repair the standard errors, not the estimator: OLS is no longer the most efficient fit under unequal variances, and the exact t laws become large-sample approximations. Section 4.6 reaches the same numbers by a different route, one that needs no variance formula at all.

4.5.3 Leverage and influence

The two plots read the residuals; two further diagnostics read each row's *weight* in the fit. Both are stated here as tools, without derivation.

Definition 4.5.1 (Leverage and Cook's distance)

Let $\mathbf{H} = \mathbb{X}(\mathbb{X}^\top\mathbb{X})^{-1}\mathbb{X}^\top$ be the *hat matrix*, which maps observed to fitted responses ($\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$). The *leverage* of row i is the diagonal entry h_{ii}, and its *Cook's distance* is

$$D_i = \frac{r_i^2\, h_{ii}}{(p+1)\,\hat{\sigma}^2\,(1-h_{ii})^2}, \qquad r_i = y_i - \hat{y}_i. \tag{47}$$

The leverage measures how strongly observation i's own response pulls its own fitted value. Leverages lie in $[0, 1]$ and sum to $p + 1$, so their average is $(p + 1)/n$; a row with a leverage several times that average sits in an unusual corner of covariate space and can tilt the fit by itself. Leverage is potential rather than harm: Cook's distance combines an unusual position with a poorly fitted response, and measures how far the whole coefficient vector moves when row i is deleted. Rows with D_i near or above 1 change the fitted equation single-handedly and deserve individual inspection: a data-entry error, an exceptional customer, or a reason to doubt the model. Figure 4.3 shows both tools catching exactly such a row, planted in the regulars' file: one customer whose 14 visits were entered as 41, the two digits transposed. Its leverage is 0.46 against an average of 0.005, it single-handedly halves the fitted slope, and its Cook's distance of 4.0 towers over every honest row's, none of which exceeds 0.03.

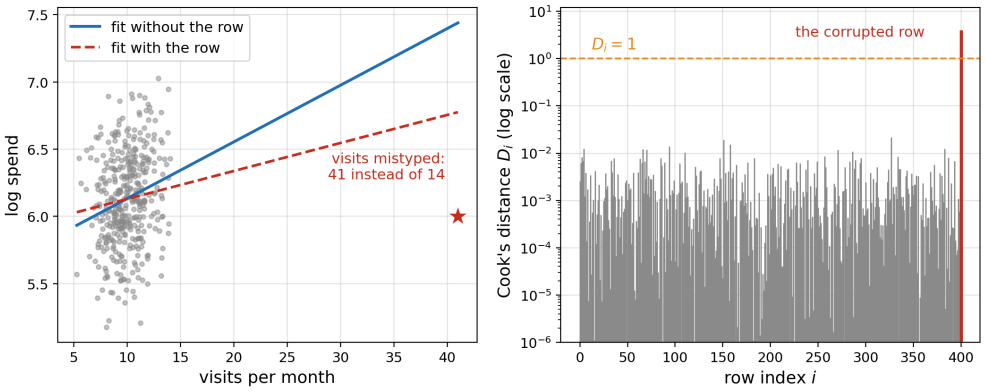


Figure 4.3: Influence diagnostics on the regulars' file with one corrupted row: 14 visits entered as 41, digits transposed. Left: the single red customer drags the fitted slope from 0.042 down to 0.021. Right: Cook's distance for all 401 rows (log scale); the honest rows stay below 0.03 while the corrupted one reaches $D = 4.0$, far past the $D_i = 1$ alarm line.

4.5.4 The assumption no plot can check

One violation is conspicuously absent from what the diagnostics can reveal: the omitted-variable failure of (A1). The residuals are orthogonal to every included column by construction; the normal equations state exactly $\mathbb{X}^\top(\mathbf{y} - \mathbb{X}\hat{\boldsymbol{\beta}}) = 0$. No plot of residuals against fitted values, against a covariate, or against anything computed from the design can therefore show that an absent variable is correlated with an included one. A fit biased by an omitted variable produces residuals exactly as healthy-looking as an unbiased one's. The most dangerous assumption is therefore the one the diagnostics cannot see: clean plots rule out curvature, heteroscedasticity and heavy tails, and say nothing at all about confounding, which is also why the causal warning of section 4.4 cannot be discharged by any amount of plotting. Independence (A2) shares this invisibility: correlated rows leave each residual individually ordinary, so neither plot betrays them either. The closing table of the chapter records which violations the diagnostics can and cannot see.

Diagnostics refute but do not certify: clean plots mean the model survived this confrontation, not that it is true. And the choice of response variable is part of the model: here the effective repair of a broken assumption was not a different estimator but a better description of the quantity being modeled. Linearity (A1) is now an assumption about the log scale, and a transformation moves the assumptions rather than removing them.

4.6 The bootstrap: inference by resampling

Every inferential product of this chapter was derived from one object: the sampling distribution of the estimator, the law (43) that says how $\hat{\boldsymbol{\beta}}$ would fluctuate across repetitions of the sampling. That law rests on (A1)–(A4), and the diagnostics have just shown that these can fail: under the funnel, the covariance formula is wrong, and with it every standard error, test and interval built on it. The repairs so far were bespoke: a transformation when one exists, a robust formula quoted without proof when it does not, each fixing one failure. This section adds the general-purpose tool: a way to *see* the sampling distribution when no formula delivers it.

Resampling the sample. Ask what the sampling distribution of $\hat{\beta}_{\text{household}}$ concretely means: if the store could recruit 400 fresh customers from the same population and we refitted, the coefficient would come out different; recruit again, different again. The histogram of those hypothetical refits *is* the sampling distribution, and its standard deviation the standard error we are after. The population cannot be redrawn from, but the sample can. The *bootstrap* [1] treats the 400 observed rows as if they were the population, and redraws samples from them. Concretely, a *bootstrap replicate* draws n complete rows $(\mathbf{x}_i, y_i)$ from the file *with replacement* (some customers drawn twice or more, others not at all) and refits the regression on the redrawn file. Repeating B times gives $\hat{\boldsymbol{\beta}}^{*1}, \ldots, \hat{\boldsymbol{\beta}}^{*B}$, and the spread of the replicated coefficients estimates the sampling fluctuation:

$$\widehat{\mathrm{SE}}_{\text{boot}}(\hat{\beta}_j) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\beta}_j^{*b} - \bar{\beta}_j^{*}\right)^2}, \qquad \bar{\beta}_j^{*} = \frac{1}{B}\sum_{b=1}^{B}\hat{\beta}_j^{*b}, \tag{48}$$

while the empirical 2.5% and 97.5% quantiles of the replicates give a *percentile confidence interval* that reads the distribution's shape directly, symmetric or not. Because entire rows are resampled (the *pairs* bootstrap), each customer's response stays attached to their own covariates: no model of the noise enters anywhere. Nothing is assumed about the variance (A3) or the distribution (A4) of the errors; what the substitution does need is (A2), independent rows drawn from one population, so that redrawing rows independently imitates how the file itself arose.

Checking the raw-spend model. Run this on the raw-spend regression, the one whose funnel falsified the printed standard errors, with $B = 2000$ refits (a few seconds of computation). The

household coefficient comes out at $\widehat{\mathrm{SE}}_{\mathrm{boot}} = 5.1$ against the nominal 4.65: the formula understates the fluctuation by ten percent. Visits moves from a nominal 3.8 to a bootstrap 4.0. The robust standard errors of section 4.5.2 report 5.2 and 4.0 for the same two coefficients: the resampling recovers them almost exactly, without anyone deriving a formula. Figure 4.4 (left) shows the disagreement as distributions: the histogram of the 2000 replicates against the Gaussian law the nominal formula asserts. On the adopted log-spend model, whose residuals contradict nothing, the same check passes: nominal, robust and bootstrap standard errors land within a few percent of one another (figure 4.4, right). The raw-scale discrepancy is ten percent, not an order of magnitude. Nothing in the printed regression output signals an error of this size, and B refits are enough to check the formula against the data.

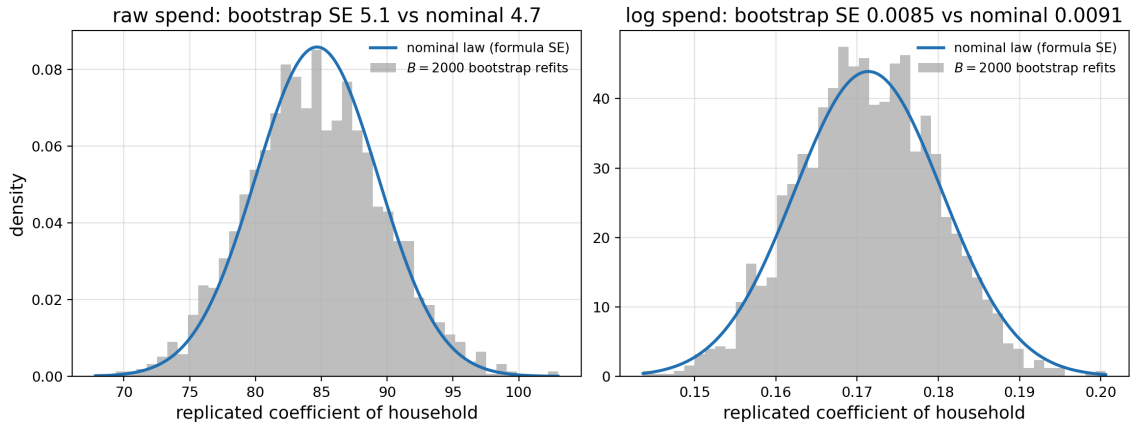


Figure 4.4: The pairs bootstrap on the regulars' regression: histogram of $B = 2000$ replicated household coefficients against the Gaussian law asserted by the nominal formula. Raw spend (left): the replicates spread wider than the formula claims, the funnel's ten-percent understatement made visible. Log spend (right): formula and replicates agree.

What the bootstrap does and does not buy. The cost is computation (B refits, trivial for least squares) plus one genuine assumption: (A2). Resampling rows independently *manufactures* independence, so when the file contains dependent rows (several customers of one household, repeated measurements), bootstrap standard errors are wrong in the same direction as the nominal ones, and the failure that no residual plot shows is one no naive resampling shows either. The other limit is subtler: the bootstrap estimates the *fluctuation* of an estimator, never its *bias*. The omitted-variable failure of (A1) resamples into a reassuringly tight histogram centered on the wrong value. Finally, the substitution of sample for population is itself an approximation that only n improves; B controls only the simulation error around it. Within those limits, the tool is remarkably general: any statistic (a coefficient, a quantile, a difference of medians) can be resampled the same way. And the device of refitting on resampled replicas returns twice in Part 2: as a sibling of chapter 5's cross-validation, and in chapter 7, where the refits on the replicas are averaged into a predictor in their own right.

4.7 Multiple tests

Encouraged, the store's data team dumps into the (log-scale) regression every column the database holds: beyond visits, household size and distance, twenty-five further recorded attributes (tenure, age, the day of the week of the first visit, and so on down the schema). Twenty-eight coefficients, one t-test each, at the customary 5%. The data being simulated, we know the truth the team does not: exactly three of the twenty-eight covariates actually influence spend.

The problem. Each single test is calibrated: *if* $\beta_j = 0$, the probability of flagging covariate j is 5%. But the team reads all twenty-eight at once, and this is exactly the many-tests situation of the previous chapter: among m true nulls, the family-wise error rate $1-(1-\alpha)^m$ compounds (46% for $m = 12$, and 72% for the $m = 25$ junk covariates of our regression). With many true nulls, at least one false rejection is therefore likely. On our data the naive screen flags four covariates: the three real ones, and the twentieth attribute, a pure noise column, at $p = 0.023$ (figure 4.5).

The corrections at work. The previous chapter built the two remedies; here they meet a regression. Bonferroni tests each coefficient at level α/m, a guarantee that needs no independence: with $m = 28$ the working threshold drops to $0.05/28 \approx 0.0018$, and real effects with mid-sized p-values are lost. Family-wise control costs power. Holm's step-down procedure spends the same budget more efficiently and rejects everything Bonferroni rejects, sometimes more. For an exploratory screen like this one, the Benjamini–Hochberg procedure controls the false discovery rate instead: sort the p-values and flag up to the last one under the rising line $k\alpha/m$.

One qualification matters more here than it did in the previous chapter. The Benjamini–Hochberg guarantee covers independent tests and certain positively dependent families, and regression coefficient tests are dependent in a specific, visible way: all twenty-eight t-statistics share the same $\hat{\sigma}$, and the components of $\hat{\boldsymbol{\beta}}$ correlate through the off-diagonal entries of $(\mathbb{X}^\top\mathbb{X})^{-1}$, the multicollinearity mechanism of section 4.4.3. Arbitrary dependence requires a modified, more conservative threshold; Bonferroni and Holm, resting on the union bound alone, are immune to the issue.

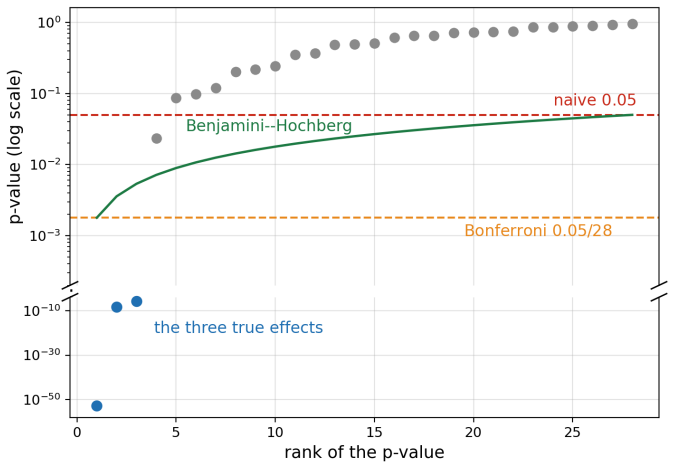


Figure 4.5: The twenty-eight sorted p-values of the full regression. The y-axis is broken: the top panel shows the threshold region, the bottom panel the three truly active covariates (blue) at their actual p-values, down to 10^{-53}. The naive 5% line admits a false discovery (the fourth point, $p = 0.023$); Bonferroni and the Benjamini–Hochberg line both flag exactly the three real effects.

On the store's regression, Bonferroni and Benjamini–Hochberg each flag exactly the three real covariates. In this example the signal is strong, so the conservative and the liberal correction happen to agree; with weaker effects Bonferroni would typically reject fewer. The previous chapter's division of labor applies unchanged (confirmatory analyses control the FWER, exploratory screens like this one the FDR), and so does its standing rule: the correction is chosen before the p-values are examined, or it controls nothing.

4.8 Conclusion

The table collects the assumptions and products of section 4.3, together with what the diagnostics of section 4.5 can and cannot see.

assumption	buys	breaks without it	in the plots?
(A1) affine mean	the fitted shape	curved fit	yes: curvature
(A1) no omitted confounder	unbiased coefficients	OVB	**no**
(A2) independent rows	error accounting	SEs too small	**no**
(A3) homoscedasticity	SE formula; Gauss–Markov	SEs wrong	yes: funnel
(A4) cond. Gaussianity	exact t/F, PIs	approximate levels	yes: QQ-plot

This chapter examined a conditional model (nothing assumed about the covariates, everything about the response given them) assumption by assumption: exogeneity bought unbiasedness, independence and homoscedasticity together bought the standard errors, Gaussianity bought exact tests and honest intervals from four hundred rows. Each product is lost when its assumption fails. This is why the diagnostics that caught the funnel, the log transform that repaired it, the resampling that estimated what the formulas could not, and the corrections for twenty-eight simultaneous tests are part of using the model, not accessories to it.

Part 1 ends here. The next chapter takes the same linear model and asks it the prologue's other question: not whether its assumptions deserve belief, but whether its predictions hold up on rows it has never seen. That question is different enough to deserve a part of its own.

References

[1] B. Efron. Bootstrap methods: another look at the jackknife. *Annals of Statistics*, 7(1):1–26, 1979.